



CHAPTER 2

Theoretical Background



There is a theory which states that if ever anyone discovers exactly what the Universe is for and why it is here, it will instantly disappear and be replaced by something even more bizarre and inexplicable.

There is another theory which states that this has already happened.

— *Douglas Adams* —



IN THIS CHAPTER, WE PROVIDE A brief account of the evolution of a star from a theoretical perspective, from its formation to post-supernova. We will discuss both the case of a single isolated star, and an isolated binary star system. It is pertinent to note that our binaries will be assumed to have formed together at birth – thus having the same age – rather than by, for example, the dynamical capture of a runaway star in a globular cluster, which introduces extra effects and considerations.

In this chapter, unless context otherwise requires, we use the following conventions surrounding the standard model and particle physics. γ refers to a photon, e^+ refers to a positron, e^- refers to an electron, ν refers to a neutrino, ν_e refers to an electron neutrino specifically, and A_ZX refers to atom X with atomic number Z and mass number A .

2.1 | Stellar Formation

*Nature and nature's laws lay hid in night:
God said, "Let Newton be!" and all was light.*

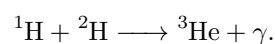
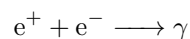
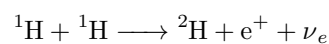
— ALEXANDER POPE

From the point of view of stellar evolution, which starts in earnest at the ZAMS, the formation of stars is interesting but ultimately has little to no impact on the nuclear evolution of the star besides setting the initial mass and composition. We provide a short recap of stellar formation here as scene-setting, but our stellar evolution models assume that a gas cloud has already collapsed and we have a star at its ZAMS.

Stars are born in clouds of gas and dust. Due to inhomogeneities in the dust formation, some regions become more dense than others. When a region of this gas with adiabatic constant γ and mean molecular weight μ reaches the so-called Jeans Mass,

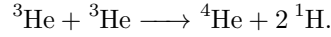
$$M_J = \left(\frac{10k_B T}{3(\gamma - 1)\mu m_H G} \right)^{\frac{3}{2}} \left(\frac{3}{4\pi\rho} \right)^{\frac{1}{2}} \sim \left(\frac{10k_B T}{G} R \right)^{\frac{3}{2}} \left(\frac{1}{M} \right)^{\frac{1}{2}}, \quad (2.1)$$

where the similarity is taking $\gamma = \frac{5}{3}, \mu = \frac{1}{2}$ for an adiabatic gas composed of H(II), then gravity becomes the dominant force in the clumps of gas. This causes a protostar to form. Generally speaking, this protostar slowly gains more mass from its surroundings (McKee & Ostriker, 2007) and, as it does so, increases in temperature. When the core temperature reaches 10^7 K, the proton-proton chain (pp -chain) begins. Bare protons are combined to form deuterium, positrons, and neutrinos. The positrons annihilate with electrons, producing photons, and the deuterium fuses with another proton to produce helium-3:

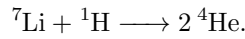
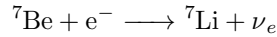
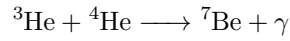


The three constituent parts of the *pp*-chain are the path 1 proton-proton chain (*pp1*-chain), the path 2 proton-proton chain (*pp2*-chain), and the path 3 proton-proton chain (*pp3*-chain). Which path the star takes is dependent on the temperature:

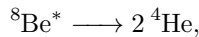
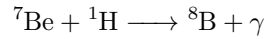
- The *pp1*-chain dominates at low temperatures, and is given by the reaction



- The *pp2*-chain dominates at temperatures $> 1.4 \times 10^7$ K, and is given by the reaction chain



- The *pp3*-chain dominates at temperatures $> 2.3 \times 10^7$ K, and is given by the reaction chain



where a superscript asterisk denotes an unstable excited state.

All three paths in the *pp*-chain ultimately produce ${}^4\text{He}$, with the key output difference being the total energy per ${}^4\text{He}$ atom; the energy yield is highest for the *pp1*-chain, and smallest for the *pp3*-chain.

The burning of ${}^2\text{H}$ – deuterium – occurs for a short fraction of the lifetime of the star, before core hydrogen burning begins in earnest. This energy release will sustain the star throughout its lifetime, providing outwards pressure resisting gravity pulling the star inwards, in a balance termed *hydrostatic equilibrium*. Once hydrogen fusion has begun, we say that the star has reached ZAMS, and is now on the Main Sequence (MS), where it will spend approximately 90% of its lifetime.

2.2 | Stellar Evolution

*It did not last: the Devil howling “Ho!
Let Einstein be!” restored the status quo.*

— “ANSWER TO POPE’S

EPITAPH FOR SIR ISAAC NEWTON”, BY J. C. SQUIRE

Now that our protostar has evolved onto the MS, it has become a star in its own right. Over the next few million years, it will continue to burn hydrogen into helium, forming a helium core. It is at this point that the evolution encounters a number of bifurcation points, depending on the mass and chemical composition of the star. In general, while the star lives, the four equations of stellar structure hold,

$$\left. \begin{aligned} \frac{dm}{dr} &= 4\pi r^2 \rho && \text{(mass continuity),} \\ \frac{dp}{dr} &= -\frac{GM\rho}{r^2} && \text{(hydrostatic equilibrium),} \\ \frac{dT}{dr} &= -\frac{3\kappa\rho L}{64\pi r^2 \sigma T^3} && \text{(radiative energy transport), and} \\ \frac{dL}{dr} &= 4\pi r^2 \rho \epsilon && \text{(energy generation),} \end{aligned} \right\} \quad (2.2)$$

where ϵ is the rate of energy eneration per unit mass, κ is the opacity of the material, σ is the Stefann-Boltzmann constant, m is the mass internal to radius r , p is pressure, ρ is density, M is total mass of the star, and L is the luminosity.

In addition to Eqn(2.2), an Equation of State (EOS) demanding $\rho = \rho(P, T)$ is desired. We take our EOS from [Pols et al. \(1995\)](#), which built upon that in [Eggleton et al. \(1973\)](#). Critically, Pols’ EOS includes the dissociation of molecular hydrogen, Coulomb interactions, and pressure ionization, which [Eggleton et al. \(1973\)](#) lacked. The EOS is an approximation to Fermi-Dirac statistics, and originates from the free energy

$$F(V, T, N_e, N_e) = -\frac{1}{3}aT^4V + kT \sum_i N_i \left(\ln \frac{N_i h^3}{\omega_i V (2\pi m_i kT)^{\frac{3}{2}}} - 1 + \frac{\chi_i}{kT} \right) + F_e(N_e, V, T) + \Delta F_e(N_e, V, T). \quad (2.3)$$

These equations represent conservation of mass, hydrostatic equilibrium, energy transport, and thermal equilibrium, respectively. Knowing constants ϵ, κ , we have four equations with four unknowns. It is usually convenient to use the *Rosseland mean* opacity ([Rosseland, 1924](#)),

$$\frac{1}{\kappa} = \frac{\pi}{acT^3} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu, \quad (2.4)$$

where a is the radiation density constant ($a = 7.57 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$) and B is the black-body intensity at frequency ν and temperature T . The function κ_ν is obtainable from atom-electron-photon interaction

(Eldridge & Tout, 2004a): bound-bound, bound-free, free-free (bremsstrahlung) transitions, and electron scattering. κ_ν is thus related to the scattering cross-section $\sigma_\nu = \int_0^{2\pi} \int_0^\pi \frac{d\sigma}{d\Omega} \sin\theta d\theta d\phi$. The Rosseland mean becomes

$$\kappa = \frac{T^5 a c^3 k_B}{\pi h^2 \int_0^\infty \frac{\nu^4}{\left(\cosh\left(\frac{h\nu}{T k_B}\right) - 1\right) \kappa_\nu} d\nu}, \quad (2.5)$$

though the integral cannot be analytically solved further. On the other hand, ϵ is obtainable from knowing the nuclear reactions occurring in the star's core, namely

$$\epsilon = \sum_i^N \text{energy output of reaction } i. \quad (2.6)$$

The reaction rates and energy outputs are typically inferred from empirical data; we take the interpolation in temperature from Caughlan & Fowler (1988), and correct the rates to approximate the cross-sections of Bahcall & Pinsonneault (1992); Bahcall et al. (1995).

All told, in the simplest sense, the internal *structure* of the star can be modelled by hand by solving Eqn(2.2) from $m = 0$ to $m = M$.

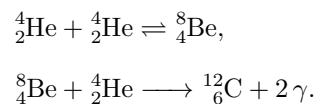
2.2.1 | The chemical composition of stars

There's a potion that can make you shrink.

— FIRST OF THE SIX IMPOSSIBLE THINGS

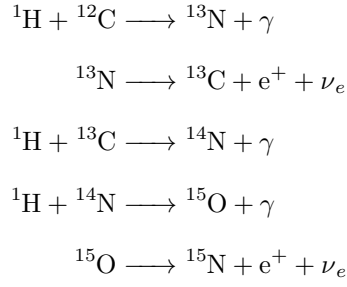
A star contains at its core a complex, multi-layered, nuclear furnace. As one atom approaches another, it must overcome the natural electrostatic repulsion so that the strong nuclear force ensures fusion occurs. This requires the bombarding atom to tunnel through the Coulomb barrier Gamow (1928), instigating fusion.

We have already discussed the *pp*-chain, fusing hydrogen into helium and up to beryllium. The two remaining parts of the story are the triple- α process and the CNO cycle. The triple- α process generates carbon, and is given by the reactions (Carroll & Ostlie, 2017)¹

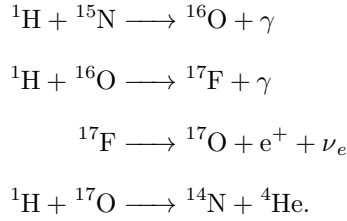


The CNO cycle generates elements up to fluorine via a branched cyclic reaction. The reactions are presented here, and are summarised in a flow chart in Figure 2.1. The start of the cycle, pre-branch, is given by

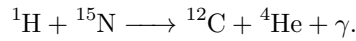
¹The reversible reaction is due to Beryllium-8 being unstable; its half-life is on the order of 1×10^{-16} s, and it decays back to $2 {}^4_2\text{He}$ if it does not react further.



Here, the reaction branches when ${}^{15}\text{N}$ undergoes a proton capture. In 1 in every 2500 captures, the reaction branches to produce (at its heaviest) fluorine, in the *NO cycle*:



In the remaining cases, the reaction instead produces carbon-12, in the conclusion of the *CN cycle*:



Further reactions essentially proceed similarly, and the core can produce up to iron before the reaction becomes endothermic. However, these elements do not dominate the star, with the exception of forming a Carbon-Oxygen core (CO core) and Oxygen-Neon core (ONe core) later in the evolution of the star. We thus group the elements into hydrogen, helium, and metals (which we define as anything with atomic number ≥ 3 , i.e., lithium and heavier elements). It is convenient to speak of the mass fraction of the element, i.e., the fraction by mass of a star that is a certain element. For our purposes, this is expressible as

$$X + Y + \sum_{i=3}^{118} \sum_j {}^i_j\text{X} = 1, \quad (2.7)$$

where X is the mass fraction of hydrogen, Y of helium, and ${}^i_j\text{X}$ as the metal² with atomic number i and mass number j . When speaking about the chemical composition of a star it is also convenient to speak of the *metallicity*, the fraction at ZAMS of the star that is metal. For this, we define the metallicity as

²It is custom in astronomy to refer to anything with an atomic number greater than 2 as a metal, rather than the conventional definition of a material conductive at absolute zero, including most of the d-block, s-block, f-block, and about half of the p-block on the periodic table.

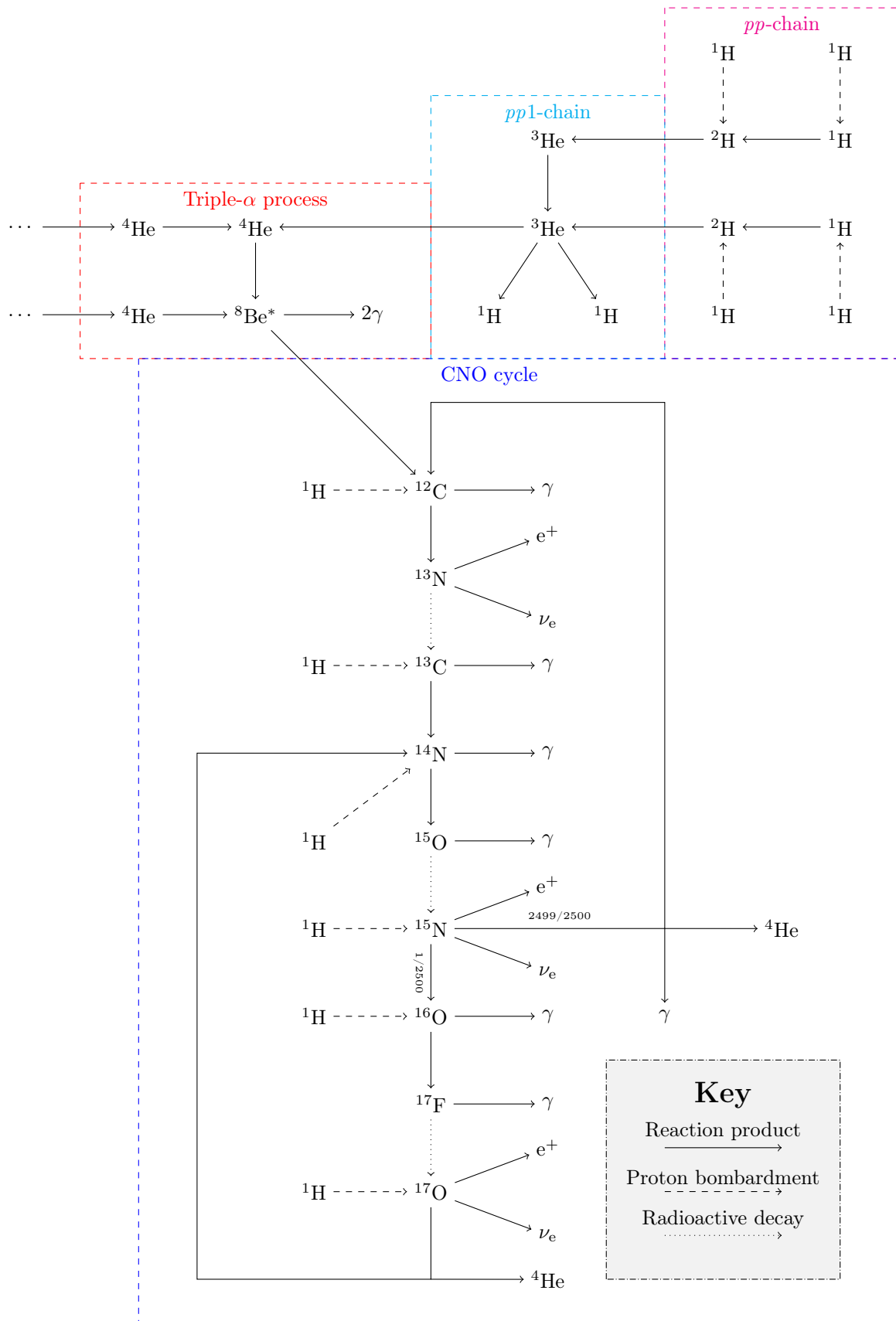


Figure 2.1: A schematic of the reactions leading up to the CNO cycle. The *pp1*-chain chain is replaced by the *pp2*-chain/*pp3*-chain chain, depending on the temperature of the star. The *pp*-chains generating the ${}^4\text{He}$ atoms on the leftmost side of the chain are omitted for simplicity. Once the ${}^{15}\text{N}$ atom has been produced, the reaction cycle is essentially self-sustaining, provided there are protons available to bombard the reactants.

$$Z \equiv \sum_{i=3}^{118} \sum_j^i X, \quad (2.8)$$

such that Eqn(2.7) becomes

$$X + Y + Z = 1. \quad (2.9)$$

Of note, abundances are not always in the literature referred to in this fashion. In some works, such as [Asplund et al. \(2021\)](#), abundances are reported in the form $\mathcal{X}_i = 12 + \log_{10}(X_i/\text{H})$. To convert between the two, the relationship

$$\mathcal{X}_i = 12 + \log_{10} \left(\frac{Z \times \frac{X_{i,\text{init}}}{56}}{0.75 - 2.5Z} \right) \quad (2.10)$$

is employed, where $X_{i,\text{init}}$ is the initial mass fraction of a given element (which we take from [Grevesse & Noels 1993](#), summarised in Table 2.1).

Element	Initial abundance	Element	Initial abundance
Hydrogen	0.7	Neon	0.099
Carbon	0.173	Magnesium	0.038
Nitrogen	0.053	Silicon	0.080
Oxygen	0.482	Iron	0.072

Table 2.1: The unweighted initial abundances by mass of our models, unless otherwise stated. Metals are then weighted by metallicity, such that, for example, the carbon abundance is $0.173 \times 0.020 = 0.00346$. The balance is assumed to be in Helium.

2.2.2 | Evolutionary Pathways

In this section, we examine the three major regimes of stellar evolution for single-stars: low-mass, intermediate-mass, and massive stars. Stars above $M_{\text{ZAMS}} \approx 80 M_{\odot}$ are out of the scope of this thesis – although they exist, and the BPASS IMF reaches $300 M_{\odot}$ ([Eldridge et al., 2017](#)), the systems of interest in future chapters are at most $\sim 80 M_{\odot}$, and thus very massive stars are out of scope for this work.

Low-Mass Stars: $0.08 M_{\odot} < M < 2 M_{\odot}$

Stars below $\approx 0.5 M_{\odot}$ do not ignite helium, and become helium WDs. Above this, they ignite helium degenerately in their cores, leading to thermal runaway. The star can peak at a luminosity close to $10^{11} L_{\odot}$ (approximately 100 times the peak luminosity of SN 1987A, per Figure 5 of [Arnett et al. \(1989\)](#)). However, this peak in luminosity is attributable to nuclear fusion in the core, and the energy does not reach the surface, let alone escape into the Interstellar Medium (ISM) – rather, the energy expands the electron-degenerate core, removing the degeneracy. During this extremely brief thermal runaway, the luminosity begins to settle and

the star is able to continue evolving. Models of low mass stars often fail at this stage and care is required to evolve them. This phenomena is called the *helium flash* and is discussed further in Section 2.6.1.

After the ignition of helium, these stars cross the so-called Hertzsprung Gap – the region between the main sequence and the Red Giant Branch (RGB) where very few stars are observed empirically.³ The phenomenon of a gap existing is attributable to the fact that stars cross this gap on their thermal timescale: whereas it spends $\sim 90\%$ of its life on the MS, it spends $\sim 2\%$ of its life crossing the Hertzsprung gap (Hoyle, 1960) as it transitions from core hydrogen burning to shell burning. After the gap, it will begin to ascend the RGB.

Intermediate-Mass Stars: $2 M_{\odot} < M < 8 M_{\odot}$

Intermediate mass stars exist above the threshold for the helium flash to occur. Helium ignition is not degenerate and the development of a helium core is not impeded. As the star ascends the Asymptotic Giant Branch (AGB), the core becomes helium-exhausted and it becomes a red giant, with a radius many hundreds or even thousands of times greater than the Sun. Its mass, sustained prior to the AGB via nuclear fusion, begins to decrease. An example of intermediate-mass evolution is shown in Figure 2.2.

Massive Stars: $8 M_{\odot} < M < 80 M_{\odot}$

Massive stars are the key focal objects of this thesis, and in many ways offer some of the most exciting phenomena we can probe. In Figure 2.3 we show the variety of evolution that this regime has, alongside the ZAMS track from $8 M_{\odot}$ to $90 M_{\odot}$. The key features are as follows. When a massive star exhausts its supply of hydrogen in its core and evolves off the MS to begin burning helium, it generally bypasses the Hertzsprung gap and evolves up the RGB. From here, stars below $\approx 15 M_{\odot}$ generally engage in a “blue loop”, where it remains at a roughly constant radius whilst increasing in temperature, before cooling at a slightly higher luminosity. This is due to excess helium being burned in shells above the core (Walmswell et al., 2015). We note that our models include blue loops, as shown in Figure 2.3 and Figure 2.4.

2.2.3 | Stellar Atmospheres

As a final note in the evolution of a single star, we must briefly address what occurs about the photosphere of the star. A star is not a planet – there is no firm, opaque boundary demarcating the inner of a star from the outer. The common definition of the “surface” of a star is the surface of last scattering (equivalently, the photosphere), which is the point radially in which the mean free path ℓ_{γ} of a photon emitted from the core of the star becomes meaningfully long (internally, $\ell_{\gamma} \sim 2 \text{ cm}$, externally, $\ell_{\gamma} \sim 4.4 \text{ pc}$).⁴ However, outflows from the star as well as the potential interaction of a partner in a common-envelope event mean that this surface is not necessarily firm: there is no sharp turn from stellar material to empty space. Thus, the simulation of stellar atmospheres is of interest to the stellar astrophysicist.

³This region is bounded (on a colour-magnitude diagram, which is a transform of the Hertzsprung-Russell Diagram) by the points $(M_v, B - V) = \{(2, 0.5), (-7, 0.1), (-6, 1.6)\}$ (Hoyle, 1960), where M_v is the V-band magnitude (magnitude in the visual band), and $B - V$ is the colour index of the star.

⁴The mean free path is a function of the particle under consideration: for a neutrino, internally $\ell_{\nu} \sim 32 \text{ pc}$ and externally $\ell_{\nu} \sim 81 \text{ Rpc} \sim 3 \times 10^{18}$ times the diameter of the observable Universe.

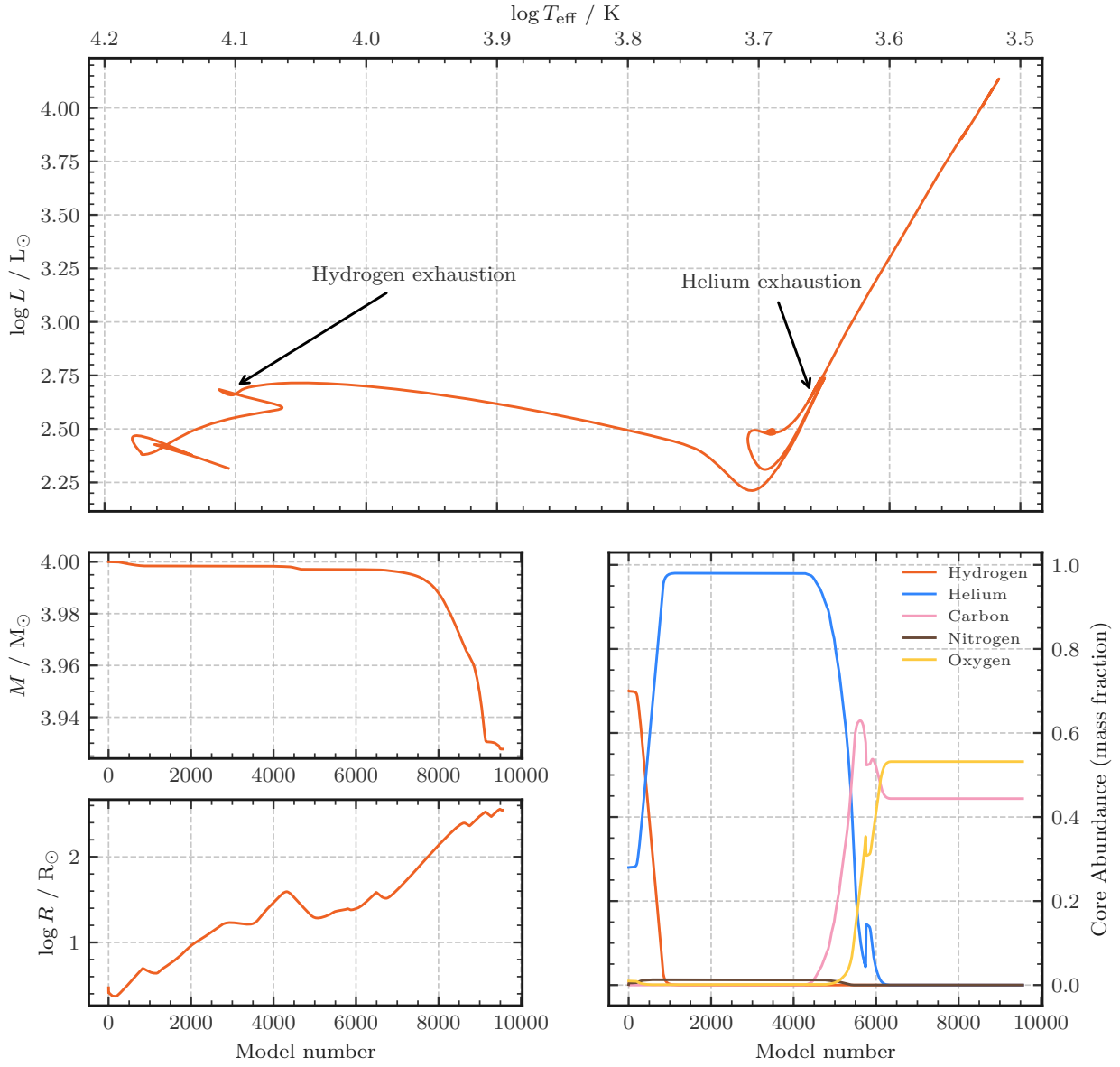


Figure 2.2: The evolution of an $4 M_{\odot}$ star from ZAMS until CO core development. The final core abundances are $Z_{\text{C}} \approx 0.44$ and $Z_{\text{O}} \approx 0.53$.

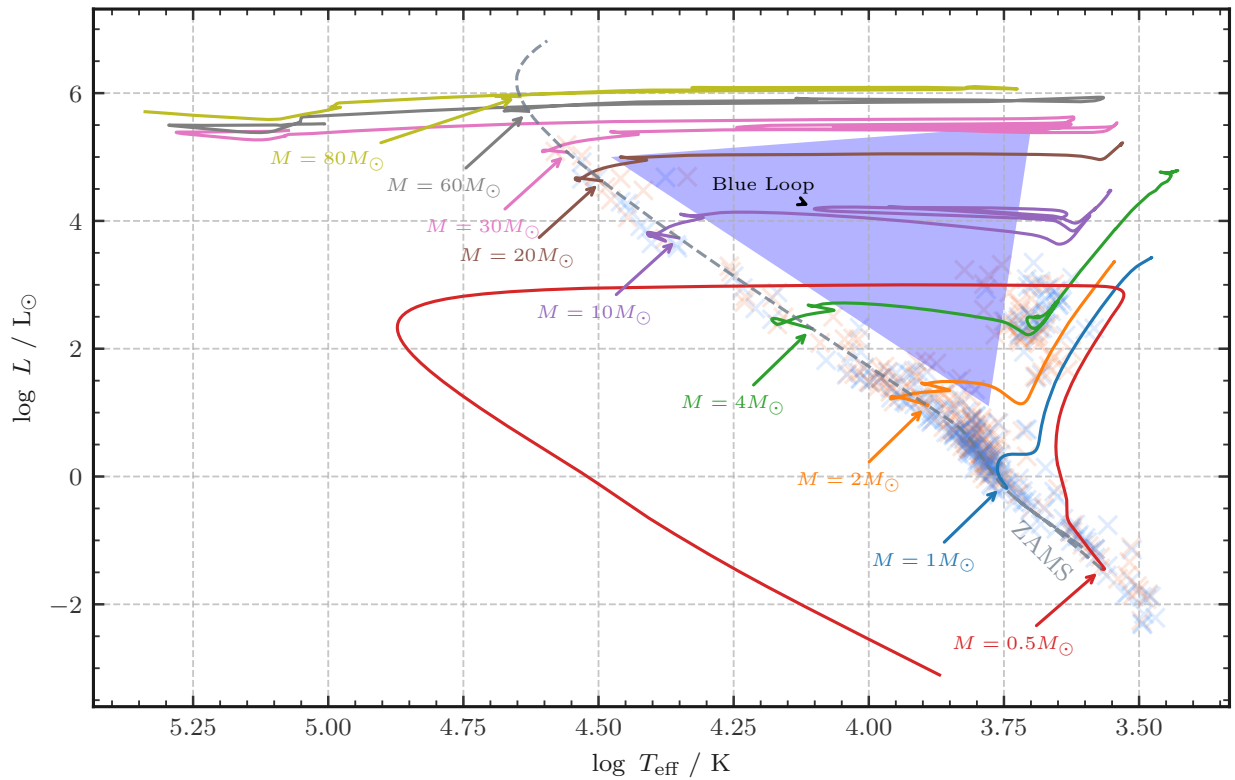


Figure 2.3: The evolution of stars in the range $0.5 - 80 M_{\odot}$. The dashed black line represents the Zero-Age Main Sequence (ZAMS), which is the origin point of all of our models, and (in this plot) extends from $0.5 M_{\odot}$ to $300 M_{\odot}$. The blue triangle is the (highly approximate) bounds of the Hertzsprung gap. Crosses indicate binaries in the Double Eclipsing Binaries catalogue (DEBCAT; [Southworth, 2015](#)), and illustrate the apparent “zone of avoidance” that defines the Hertzsprung gap. Blue crosses are primaries, and orange crosses are secondaries.

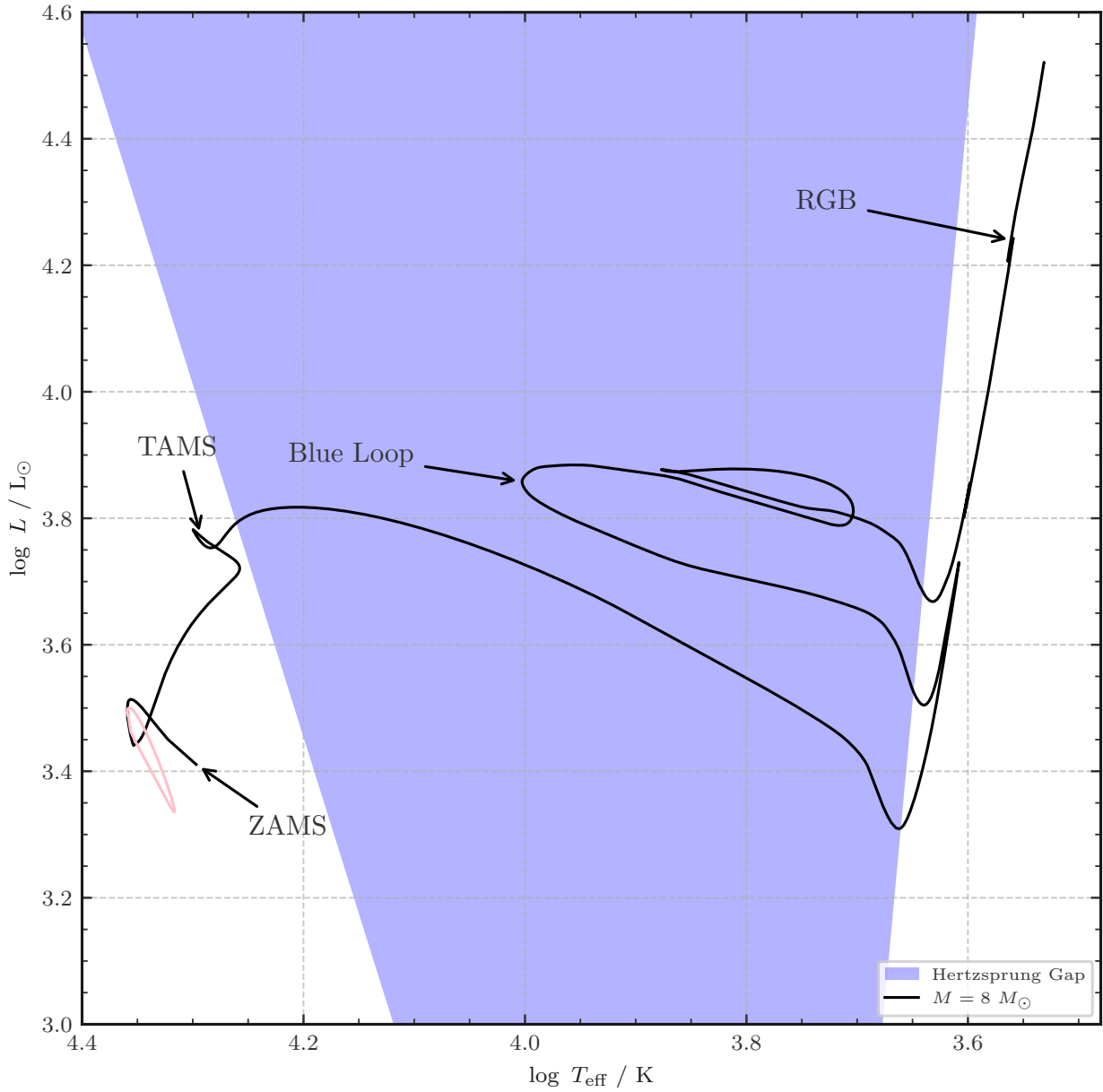


Figure 2.4: The evolution of an $8 M_{\odot}$ star. Highlighted is the star's Zero-Age Main Sequence (ZAMS), Terminal-Age Main Sequence (TAMS), the blue loop, and the Asymptotic Giant Branch (AGB). The Hertzsprung gap is also highlighted. In this particular model, the star achieves CNO equilibrium in the leftmost (hottest) region of the plot, highlighted in pink.

However, this introduces another rather large layer of complexity. The synthesis of stellar spectra is not a trivial task, even without considering the complexity of evolving the star simultaneously. [Groh et al. \(2014\)](#) is one of the very few instances of coupled spectroscopic calculations with evolutionary calculations, which only evolved a single star. Moreover, this thesis is concerned with detailed modelling of the stellar *interior*, and thus the atmosphere is an interesting sidenote, but one we shall not pursue.

2.3 | Supernovae

<i>And on the pedestal these words appear:</i>	<i>Pallida mors æquo pulsat pede pauperum</i>
<i>“My name is Ozymandias, king of kings:</i>	<i>tabernas regumque turres.</i>
<i>Look on my works, ye Mighty, and despair!”</i>	<i>Pale death knocks with the same tempo upon the</i>
<i>Nothing beside remains.</i>	<i>huts of the poor and the towers of kings.</i>
— PERCY BYSSHE SHELLEY, <i>OZYMANDIAS</i>	— HORACE

A star’s lifetime is not infinite. The key assumption underpinning stellar evolution is that of hydrostatic equilibrium: the counterplay of gravity against outwards pressure. In short, a dynamically stable, non-rotating star composed of an ideal gas with adiabatic constant γ and gravitational potential energy Ω radiates energy via

$$L = -\dot{E} = -\frac{3\gamma - 4}{3(\gamma - 1)} \frac{d\Omega}{dt} - \frac{d\gamma}{dt} \left(\frac{1}{(\gamma - 1)^2} \right), \quad (2.11)$$

If the chemical gradient is largely unchanged, $\frac{d\gamma}{dt} \sim 0$ so the luminosity reduces to

$$L = -\frac{3\gamma - 4}{3(\gamma - 1)} \frac{d\Omega}{dt} \quad \text{where} \quad U = -\frac{1}{3(\gamma - 1)} \Omega, \quad (2.12)$$

where U is the total internal energy of the star. Since $\dot{E} < 0$ for a star radiating energy, and $\Omega < 0$ by definition (since assembling a star is energetically favourable), we have that $\dot{\Omega} < 0$ and $\dot{U} > 0$, i.e., that as the star radiates energy it attempts to shrink. This shrinking is arrested by the nuclear fusion in the core providing outwards pressure, meaning the star is stable and resists explosion. As the star continues to burn through its fuel supply and eventually exhaust it, gravity will win as the dominant force, and result in the star imploding in a CCSN. From a thermodynamic perspective, as the star evolves, it shrinks as energy is bled off into the interstellar medium. This shrinkage compresses and heats the core, meaning the star’s specific heat capacity is negative. This system is thermodynamically unstable, and thus every star either collapses into a remnant (cf. BH, NS) or is sustained in some form by electron degeneracy pressure (cf. WD).

In any case, the breaking of hydrostatic equilibrium results generally results in⁵ a supernova, which can be classified into subtypes, as shown in Figure 2.5. In this event, gravity pulls the stellar core inwards, before

⁵The extent of the veracity of this statement is an area of active research: [Reynolds et al. \(2015\)](#) presents a short summary.

“bouncing” and ejecting the envelope into the interstellar medium. The physics of this approximately second-long event is highly uncertain, and is in general unable to be modelled by detailed stellar evolution codes. All models we generate – assuming they have not been terminated earlier – will terminate at the moment of supernova.

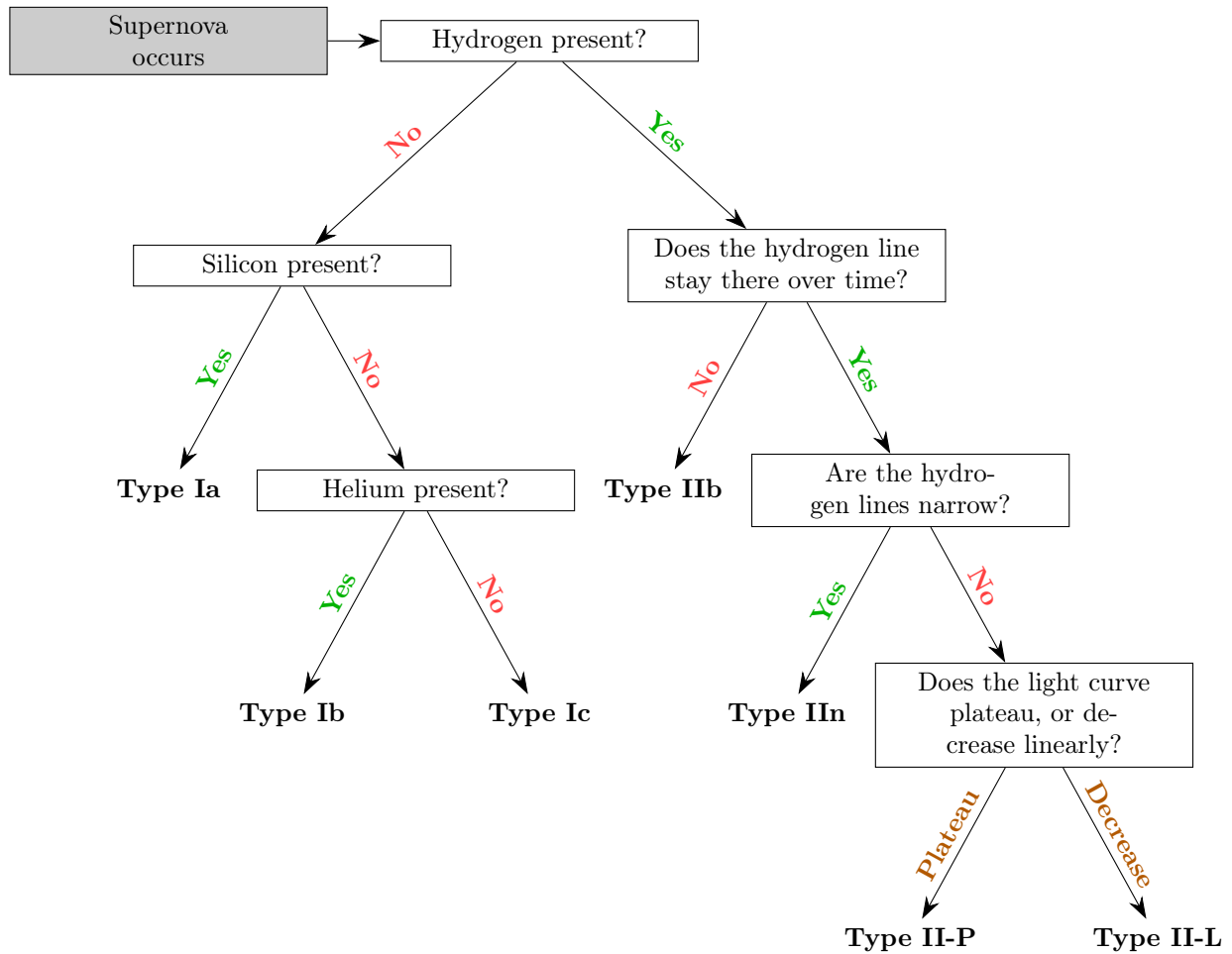


Figure 2.5: A rough outline of supernova classification. One can see that the presence of hydrogen lines distinguishes a Type I from Type II supernova. Type II supernova subtypes are determined by the behaviour of their light curves – how the apparent magnitude changes against time – and thus are determined by the luminosity of the supernova, something detailed models are not able to model. On the other hand, the presence of hydrogen/silicon/helium in the supernova is something that detailed codes *are* able to model. There are further delineations into Type Ia/b/c-pec and Type IIpec, denoting peculiar supernovae, however these are rare and are in some sense exception categories, where the behaviour does not fit nicely into a taxonomy tree.

Importantly for the nucleosynthesis of iron-group and heavier elements (which cannot be formed via hydrostatic nuclear fusion,⁶ as shown in Figure 2.6), the supernova shockwave propagating through the core causes a new phase of nucleosynthesis, albeit briefly.

2.3.1 | Type Ia supernovae

There is another class of supernovae we consider in this thesis, namely type Ia supernovae. These are formed in two major channels:

⁶Strictly speaking, fusion can occur for elements heavier than Iron-56, but the reaction is endothermic; it requires more energy than is released, meaning it is unsustainable.

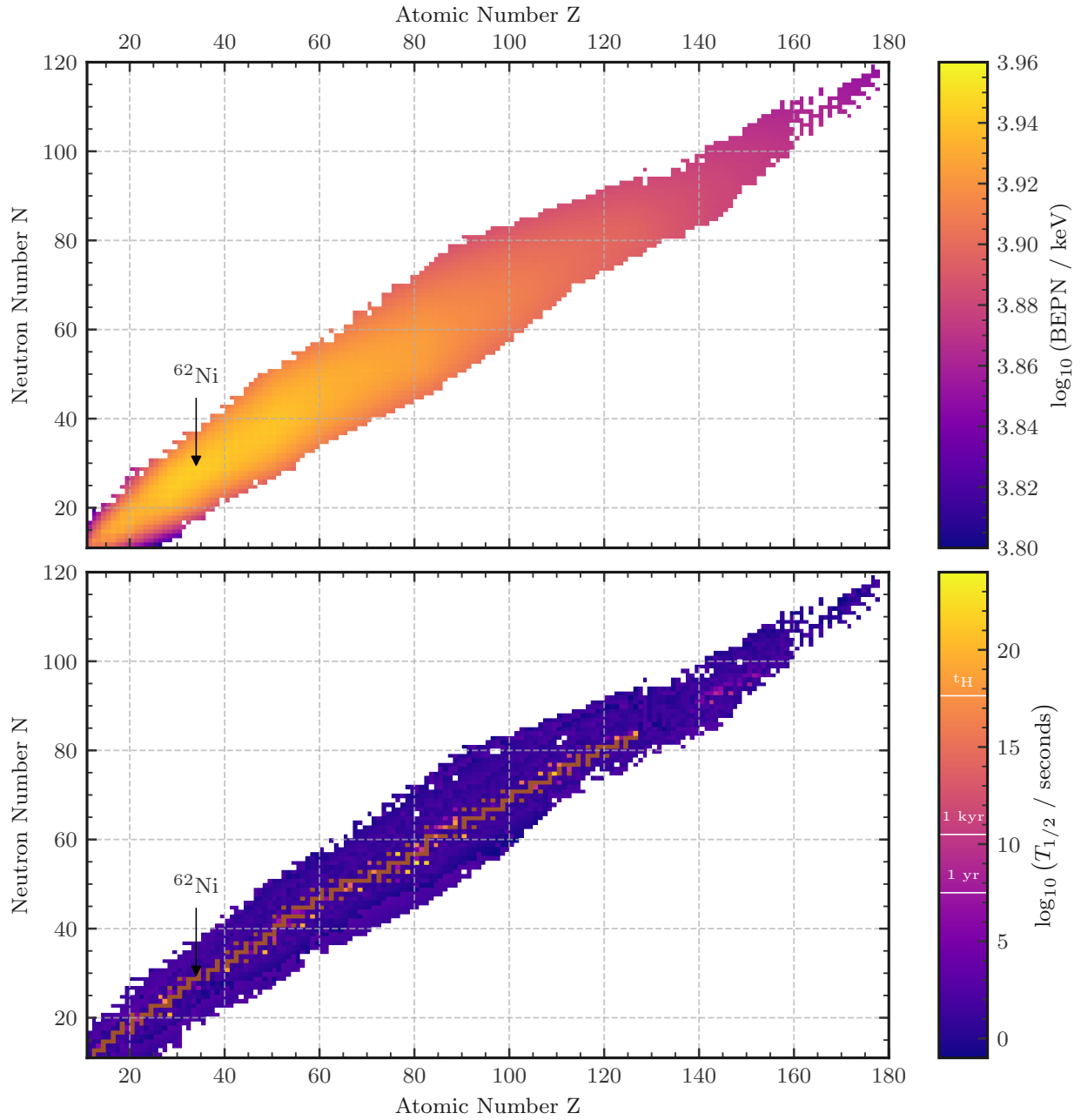


Figure 2.6: *Top:* The binding energy per nucleon and half-lives for elements up to Oganesson. The most stable element of fusion here is Nickel-62. Elements lighter than Nickel-62 will fuse towards it, and elements heavier will undergo fission to reach it. *Bottom:* The half-life in seconds across the same isotopes. Marked on the colorbar axis are 1 yr, 1000 yr, and $t_H = 13.8 \text{ Gyr}$. Stable elements are marked in brown. Underlying data for both plots are from the International Atomic Energy Agency.

1. When a WD star exceeds the maximum mass by which electron degeneracy pressure can sustain the star. This mass is termed the Chandrasekhar mass, $M_{\text{ch}} \approx 1.44 M_{\odot}$. When material accretes onto a white dwarf, its mass will increase, and if the accretion does not stop, the mass will exceed the Chandrasekhar mass. This means electron degeneracy pressure is no longer able to sustain the star, and the core collapses further into a Type Ia (or thermonuclear) supernova.
2. During the merger of a CO WD binary in which a super-Chandrasekhar WD or potentially helium- or carbon-burning star is produced [Webbink \(1984\)](#); [Ruiter & Seitenzahl \(2025\)](#).

For a more thorough discussion on type Ia supernovae, the reader is referred to [Ruiter & Seitenzahl \(2025\)](#) for a review.

2.4 | Two to tango: Binary Evolution

Introducing a second star into the mix complicates the evolution. Whereas before a star losing mass (through, for example, stellar winds) was assumed to eject it into the ISM, it may now accrete that material directly onto its companion. In addition, either star may in principle increase in radius enough to overlap with its companion, or even touch surfaces – introducing new dynamics into the mix.

The most marked change for a binary compared to a single star is that its trajectory on the Hertzsprung-Russell Diagram (HRD) is altered. The MS is changed somewhat, but (for longer than ~ 5 d periods) the interaction takes place post-MS and is shown in Figure 2.7.

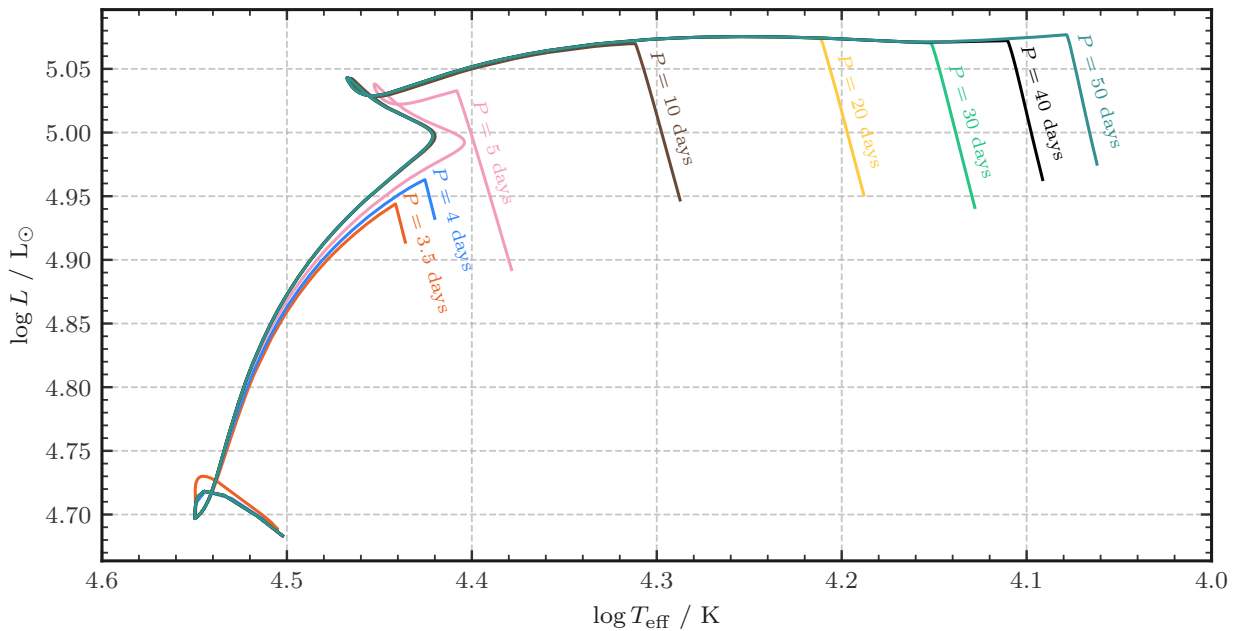


Figure 2.7: The HRD for an $21 M_{\odot}$ star with an (unshown) $10 M_{\odot}$ companion with varying period. In this figure, only the initial binary period was free to change between runs. Short periods interact closer to the MS. The evolution was terminated 1,000 models after the primary star achieved RLOF for the first time.

2.4.1 | Binary star kinematics

There are several formation channels for a binary star system. Binaries may be formed from birth (as discussed in Section 2.1, with the caveat that in this scenario two clumps of gas form near each other), via fragmentation of a protostar's disc, dynamical capture of one star by another (comparable to one of the suggested origins of Phobos and Deimos, per [Burns \(1978\)](#)), or via a higher multiplicity system ejecting one or more of its constituent stars, perhaps via a natal supernova kick.

In any case, once formed and the orbit is stabilised, binary systems obey Kepler's three laws. In the generalised binary regime, however, the assumption that $M_1 \gg M_2$ (which underpins most statements of Kepler's laws) is not necessarily true – for instance, the system LSS 3074 has two stellar components with ZAMS masses of approximately $30 M_\odot$ and $10 M_\odot$ ([Raucq et al., 2017](#)). Moreover, the fact that there are two stars in orbit rather than one object orbiting another means that Kepler's laws must be stated more precisely. Formally, they can be expressed as (see Appendix A for derivations):

1. All orbits are ellipses, with both stars orbiting a common barycentre at the focus of the ellipse,
2. A line segment joining the two stars sweeps out equal areas in equal times,
3. The square of the period of the system is proportional to the cube of the semimajor axis of the ellipse (the binary separation), with the proportionality term dependent on the sum of the masses.

The third law can be stated more precisely as

$$T^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3, \quad (2.13)$$

where T is the period of the binary, M_1 and M_2 the constituent masses, $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the universal gravitational constant, and a is the semimajor axis of the binary. A schematic of the orbit of a binary system is given in Figure 2.8.

2.4.2 | Roche Lobes

A binary system is complicated by the presence of its companions in several different ways. As alluded to in the previous section, the presence of a companion warps space-time in such a way that material can be quite distorted from a sphere or oblate spheroid but still remain gravitationally bound to the system. In the extremal case, a star can grow to fill or even exceed its Roche Lobe, which is the tear-drop shaped equipotential surface within which material remains gravitationally bound to its star. In Figure 2.9 we display these equipotential surfaces for a range of different mass ratios. It is pertinent to note that the shape of all of the Roche Lobes change rapidly at low mass ratios, however at higher mass ratios the shapes change less rapidly. In an equal-mass ratio binary, the binary configuration has rotational symmetry about the $\mathcal{L}_1$ point, and reflectional symmetry about the x -axis and the line $y = \mathcal{L}_1$.

There are two primary Roche Lobe points we consider: the $\mathcal{L}_1$ point and the $\mathcal{L}_2$ point. The $\mathcal{L}_1$ point (also known as the inner Lagrangian point) exists on the line connecting the centres of the two stars, the

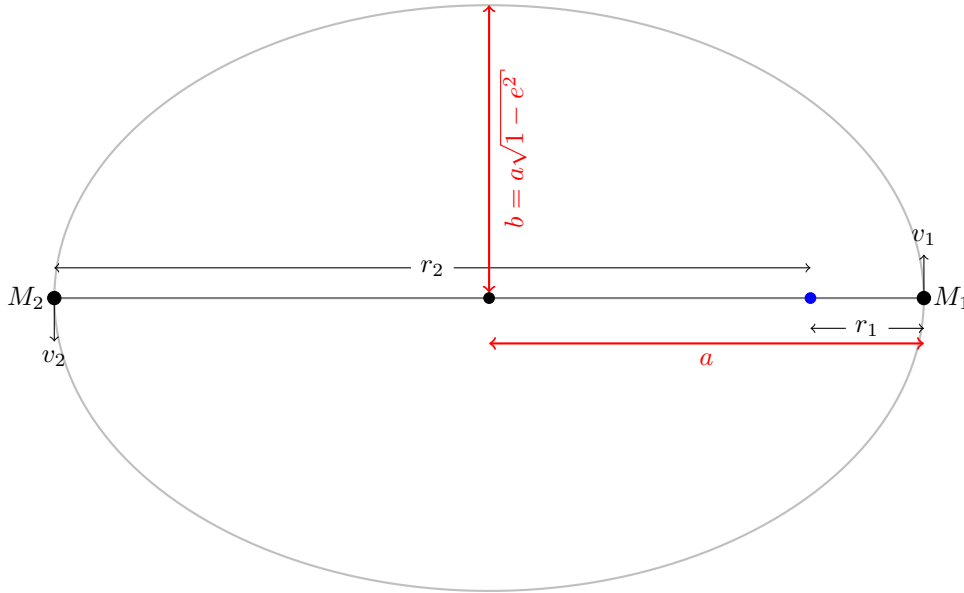


Figure 2.8: A schematic of a simple binary star system with eccentricity e , semimajor axis a , and semiminor axis b . The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a smaller distance r_1 to the barycentre by Kepler’s first and Newton’s second laws. The elliptical orbital trajectory is shown in grey. Figure adapted from [Richards \(2021\)](#).

$\mathcal{L}_2$ point (the outer Lagrangian point) is located on the other side of the secondary to the inner point. The dimensionless x -ordinate of the $\mathcal{L}_1$ point, ξ , measured in terms of the separation a , is given as the solution to

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}, \quad (2.14)$$

where $\xi_1 = \frac{M_2}{M_1+M_2}$, and $\xi_2 = \frac{M_1}{M_1+M_2}$. This expression is derived in Appendix A

This is a quintic expression in ξ , and solving this results in the real ξ -ordinates of five extrema. The extrema are labelled $\mathcal{L}_1$ through $\mathcal{L}_5$. $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3$ are saddle points of the potential, and $\mathcal{L}_4$ and $\mathcal{L}_5$ are maxima. No analytical function for ξ exists unless either or both of ξ_1, ξ_2 are zero (which would only be the case for a single star, and thus the Roche lobe would not exist in the first instance), thus we must make approximations.

The $\mathcal{L}_1$ point

The inner Lagrangian point has the approximation from [Eggleton \(1983\)](#), which is

$$\frac{R_L}{a} \approx \frac{0.49q^{\frac{2}{3}}}{0.6q^{\frac{2}{3}} + \ln\left(1 + q^{\frac{1}{3}}\right)}. \quad (2.15)$$

We show the circle with this radius R_L in Figure 2.10 plotted against the exact solution for $\eta = \mathcal{L}_1$. Eggleton’s formula is a volume-equivalent one; although his radius $R_L < \xi$, the volume of an Eggleton-Roche sphere $\frac{4}{3}\pi R_L^3$ is approximately equal (to within 1 per cent) of the numerically integrated volume, as is shown in Figure 2.11.

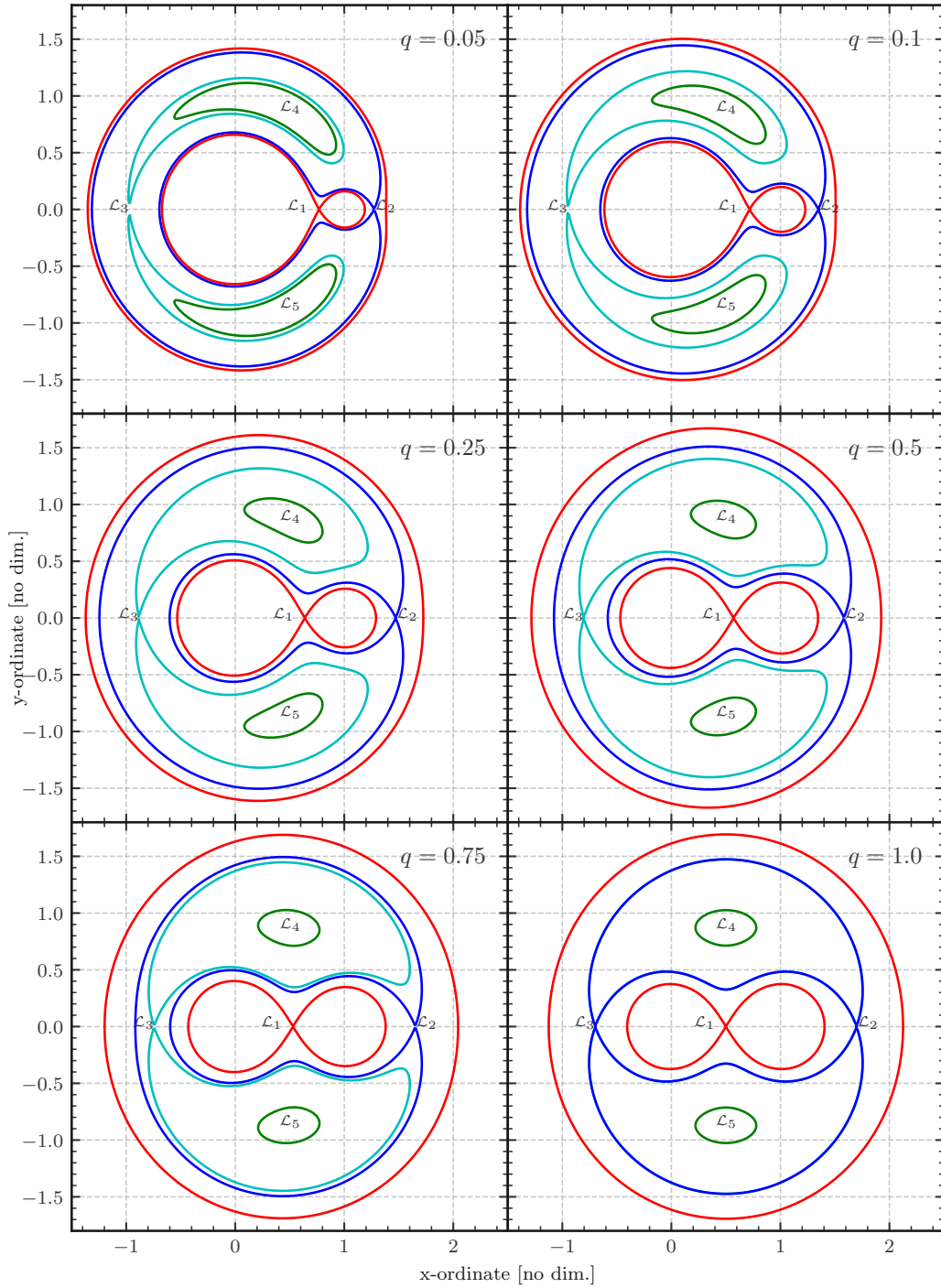


Figure 2.9: The dimensionless Roche potential as a function of mass ratio q . Note that the shape of the Roche potential changes markedly at low mass ratio, and changes more slowly as the mass ratio approaches unity. Distances have been normalised such that the primary star is at $(x, y, z) = (0, 0, 0)$ and the secondary star is at $(x, y, z) = (1, 0, 0)$, i.e., we measure distances in units of separation and that the coordinate system is centered on the primary star.

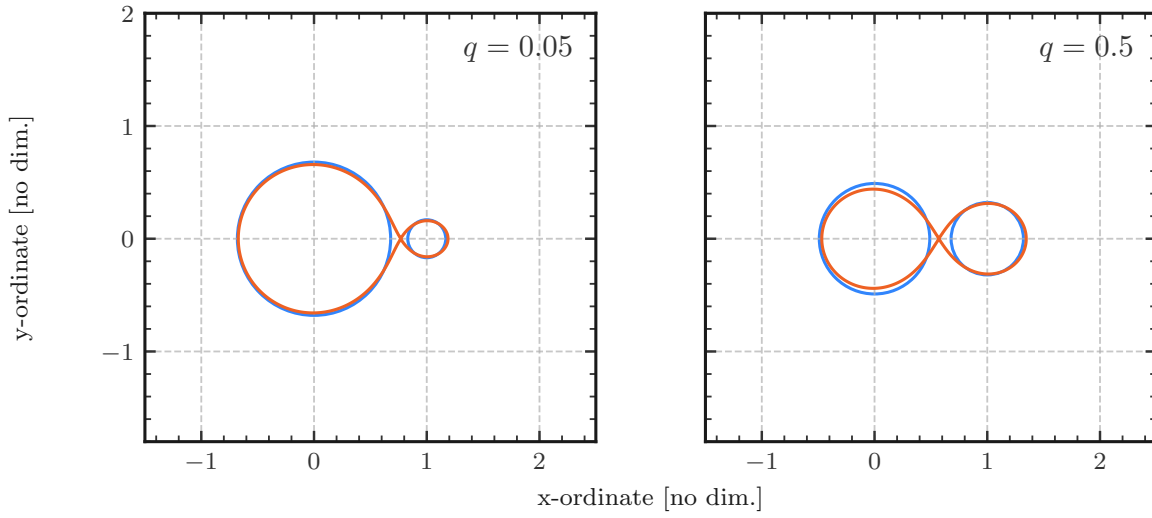


Figure 2.10: A schematic of the Roche lobe formula Eqn(2.15). Here, the exact solution to the Roche lobe formula ξ in Eqn(2.14) (orange) vs the fit provided by Eggleton (1983) (blue). Although precise details of the Roche lobe, such as the apex of the teardrop at the point ξ , are lost, the volume is preserved to within 1 per cent accuracy over all mass ratios.

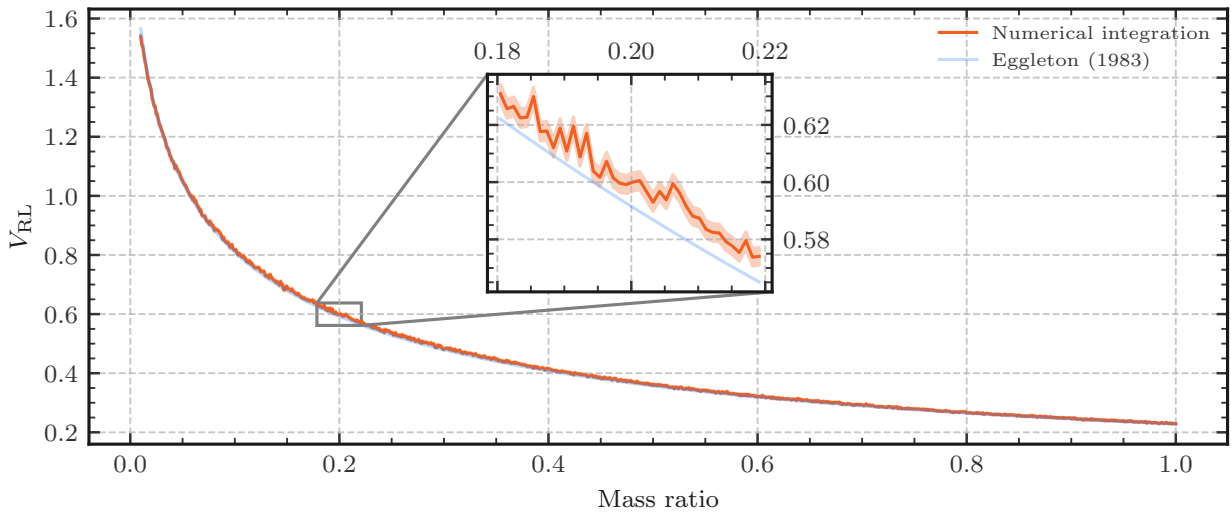


Figure 2.11: The volume of the Roche Lobe computed numerically through Monte-Carlo integration (orange) vs the volume assuming Eqn(2.15) (blue). The Eggleton approximation fits well to the volume, with around 1 per cent error across the entire mass ratio range (truncated here to $q \leq 1$). The orange band is the uncertainty range as computed by a Monte-Carlo integrator from pyAstronomy (Czesla et al., 2019), and is a binomial standard deviation.

The $\mathcal{L}_2$ point

The outer Lagrangian point, $\mathcal{L}_2$, is given by a different zero of the potential function. The approximation we use is that of [Marchant et al. \(2016\)](#), where

$$R_{L2} = R_L(q) \left(1 + \frac{0.299 \arctan(1.84q^{-0.397})}{q^{0.520}} \right), \quad (2.16)$$

and $R_L(q)$ is given by Eqn(2.15).

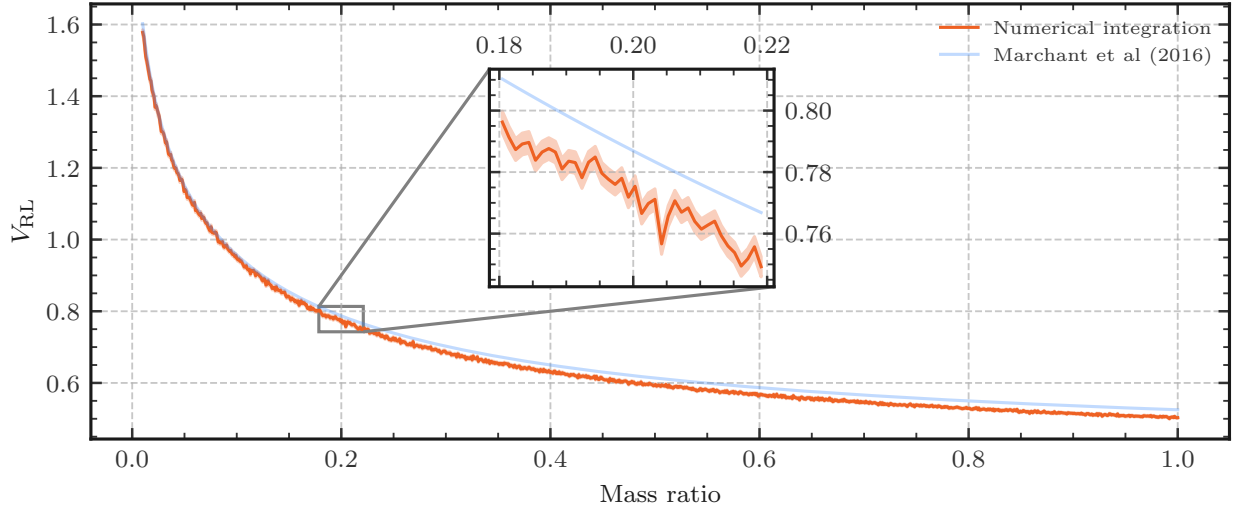


Figure 2.12: As for Figure 2.11, but for the $\mathcal{L}_2$ point with formula given by Eqn(2.16) ([Marchant et al., 2016](#)).

We note that [Marchant et al. \(2016\)](#) uses an inverse mass ratio definition compared to [Eggleton \(1983\)](#), that is, [Marchant et al. \(2016\)](#) takes $q_{\text{Marchant}} = \frac{M_1}{M_2}$ whereas [Eggleton \(1983\)](#) takes $q_{\text{Eggleton}} = \frac{M_2}{M_1} = \frac{1}{q_{\text{Marchant}}}$. This is accounted for in Eqn(2.16) such that the mass ratios are consistent (and we take $q = q_{\text{Eggleton}}$), but other texts do not necessarily differentiate or describe the difference.

2.4.3 | Mass Transfer

And a cake that can make you grow.

— SECOND OF THE SIX IMPOSSIBLE THINGS

When two stars are sufficiently close in proximity, surface and near-surface events can cause accretion of material from one star to another due to the primary's stellar winds (whereas in the single-star case, this would be matter lost to the ISM). The nature of mass transfer is a complicated affair; most binary star population synthesis codes typically inherit their mass loss prescriptions from papers such as [Bondi & Hoyle \(1944\)](#); [Hurley et al. \(2002\)](#), and this approach is what we detail in this section.

Mass transfer can broadly be sectioned into four main categories:

1. $\dot{M}_{\text{wind}}$, where matter is lost through stellar winds,

2. $\dot{M}_{\text{RLOF}}$, where matter is transferred through one (or more) of the Lagrange points in a binary (cf. Section 2.4.2),
3. $\dot{M}_{\text{acc}}$, where matter is transferred through accretion (or interaction with a disk), and
4. $\dot{M}_{\text{Bondi}}$, where matter proceeds through a [Bondi & Hoyle \(1944\)](#) type mechanism.

Our mass loss rate for each star is constructed from these four terms as

$$\dot{M} = \dot{M}_{\text{wind}} - \dot{M}_{\text{Bondi}} + \dot{M}_{\text{RLOF}} - \dot{M}_{\text{acc}}. \quad (2.17)$$

Wind mass-loss

Wind mass loss computations are (generally) performed using a mix of theoretical and empirical mass-loss rate prescriptions of [de Jager et al. \(1988\)](#) and [Vink et al. \(2001\)](#). The precise scheme is detailed in [Eldridge et al. \(2017\)](#).⁷ There are two critical modifications we make to their scheme.

The first modification surrounds the transition into the Wolf Rayet (WR) regime. Traditionally this has been a sharp transition when the surface hydrogen mass fraction becomes less than 0.4 and the surface temperature is above 10^4 K. This is a reasonable approximation, though if a star reaches this phase and is *not* yet a WR star, the code will fail. To account for this we gradually turn on these enhanced WR rates when this condition approaches, smoothing the transition.

The second modification we have made is that if a star has a hydrogen exhausted core and begins approaching the WR temperature range, we assume that WR mass loss must begin and thus turn on the slowly-turning on WR regime.

Mathematically, the computation for $\dot{M}_{\text{wind}}$ is more involved, though follows the same essential structure as [Vink et al. \(2001\)](#). Details are provided in Section A.4.

Roche Lobe overflow (RLOF)

Matter extruded from the Lagrangian points is assumed to transfer with a mass loss rate similar to [Hurley et al. \(2002\)](#). We take

$$\dot{M}_{\text{RLOF},i} = k \cdot \max \left(\ln \left(\frac{R_i}{R_{\text{RL},i}} \right), 0 \right)^3 \left(\frac{M_i}{M_{\odot}} \right)^2 M_{\odot} \text{ yr}^{-1}. \quad (2.18)$$

Here, k (referred to in our code as RMT) parameterises effectively how much of the star we permit to exist outside of its Roche lobe before it is accreted – i.e., the voracity of RLOF. The choice of this can be somewhat arbitrary. [Hurley et al. \(2002\)](#) for example took $k = 3 \times 10^{-6} \min \{M_1, 5\}^2$, where the coefficient was “chosen by experiment to ensure that the mass transfer is steady”, and warned that for degenerate stars a more appropriate choice would be to increase this by a factor of 10^3 , corroborated by [Claeys et al. \(2014\)](#).

⁷In other codes such as MESA, this is referred to as the “Dutch” mass loss prescription owing to the nationality of the authors of [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#).

We choose to adopt $k = 3 \times 10^{-6} \times 10^3 = 3 \times 10^{-3}$, with no dependence on the mass. This permits the star to overfill its Roche lobe even in tight binaries, but still allows efficient mass transfer.

Matter exceeding $\mathcal{L}_2$ (the outer Lagrange point) does not currently receive special treatment. The implications of $\mathcal{L}_2$ overflow are significant. This ejection will lead to unstable mass transfer, shrinking the orbit and likely inducing a Common Envelope Evolution (CEE) event (Fragos et al., 2023). There is debate about the appropriate treatment here. Linial & Sari (2017) indexes the $\mathcal{L}_2$ rate to the $\mathcal{L}_1$ rate via an efficiency parameter. Li & Li (2013) considers the case analogous to $\mathcal{L}_1$ outflow, where material outflowing from $\mathcal{L}_2$ acts to retard the angular momentum. In their notation $\frac{dJ}{dt} = -\epsilon \dot{M}_1 a_{L2}^2 \omega$ (compare $\frac{dJ}{dt} = -\beta \dot{M}_1 a_{L1}^2 \omega$ for RLOF through $\mathcal{L}_1$) for controllable parameters β and ϵ . The ambiguity of treatment leads to some codes to terminate at the onset of $\mathcal{L}_2$ outflow, such as POSYDON (Fragos et al., 2023). $\mathcal{L}_2$ outflow requires stars to be in reasonably tight orbits (~ 2 d), which we will approach, but none of our stars will enter this regime, and thus we do not afford it special treatment.

Accretion

Our wind accretion and RLOF accretion rates are taken from Bondi & Hoyle (1944); Hurley et al. (2002), and limits the accretion rate by the thermal timescale of the star. The prescription is that

$$\dot{M}_{\text{acc}} = \min \left(\dot{M}_{\text{RLOF}}, \frac{M}{T_{\text{KH}}} \max \left(1 - \frac{\omega}{0.8\omega_{\text{crit}}}, 0 \right) \right), \quad (2.19)$$

where ω and ω_{crit} are the spin rate and critical spin rate of the star. We generally assume that the star is composed of individually rotating shells of variable thickness.⁸ If N is the number of meshpoints that the star is composed of (for our cases, normally $N = 199$), then the moment of inertia is given by

$$I = \frac{2}{5} M_c R_c^2 + \sum_{k=1}^N \frac{2}{3} (m_k - m_{k+1}) r_k^2, \quad (2.20)$$

where m_k is the mass internal to meshpoint k (similar for r_k), and a subscript of c refers to the central meshpoint. It is assumed that the central meshpoint sits atop a solid sphere, hence $\frac{2}{5}$ as its prefactor.

Bondi-Hoyle wind accretion

Bondi-Hoyle accretion for a circular orbit is given by

$$\dot{M}_{\text{Bondi}} = \min \left[- \left(\frac{GM_2}{v_W^2} \right)^2 \frac{\alpha_W}{2a^2} \frac{1}{(1+v^2)^{3/2}}, 0.8 \right] \dot{M}_{\text{wind}}, \quad (2.21)$$

where we have assumed that the orbital velocity is given by

$$v^2 = \frac{G(M_1 + M_2)}{av_W^2}, \quad (2.22)$$

and that the wind velocity is given by

⁸Also known as shellular rotation.

$$v_W = 2\beta_W \frac{GM_1}{R_1}. \quad (2.23)$$

Generalising this to an eccentric orbit causes the first argument in Eqn(2.21) to pick up a coefficient of $\frac{1}{\sqrt{1-e^2}}$. There are two free parameters to select: α_W and β_W . These ostensibly depend on stellar types, though we select for simplicity $\alpha_W = \frac{3}{2}$ (Boffin & Jorissen, 1988) and $\beta_W = 7$ (Lamers et al., 1995) for O-type stars. This parameterisation is likely incomplete; Hurley et al. (2002) argues that as per Lamers et al. (1995), β_W should vary between 7 for O-type stars to 0.5 for A and F-type stars, perhaps descending as low as $\beta_W \approx \frac{1}{8}$ for cool supergiants. We leave this implementation for future work under the assumption that our mass-loss rate is correct to within an order of magnitude.

2.5 | Post-Supernova

Wit nodded.

“Oh, is this the place where I talk?” Kaladin said.

“If you wish.”

“I don’t wish. Get on with the story.”

— “RHYTHM OF WAR”, BY BRANDON SANDERSON

After a supernova, the star leaves behind its core as a compact remnant. Generally, these are either black holes or neutron stars. The interior of a neutron star is reasonably complex, with superfluid neutrons within the crust (Lattimer & Prakash, 2001) and a latticed core-crust interface (Rueda et al., 2014), with “pasta phases” at the interface (Zdunik et al., 2017). The interior of a black hole is impossible to observationally probe without violating causality.

The supernova ejects some fraction of material M_{ej} , leaving behind a remnant of mass M_{rem} . Eldridge & Tout (2004b) gives a relationship between ejecta, remnant, and stellar masses by assuming a supernova releases 10^{46} J ($=10^{53}$ erg) of energy, of which 1 per cent is transferred into the envelope. The full relation between the binding energy U , the stellar mass M_* , and the ejecta and remnant masses is thus

$$\int_{M_{\text{rem}}}^{M_*} \left(U - \frac{GM}{r(M)} \right) dM = 10^{44} \text{ J}, \quad (2.24)$$

where $r(M)$ is the radius of the sphere containing mass M of the star. The post-supernova geometry is given by Tauris & Takens (1998).

The orbital motions of a binary system post-secondary supernova are now deterministic from four initial parameters: m_1, m_2, a , and e . Given their high masses and small radii, we may approximate them as point masses, and thus their motion is now wholly determined by the emission of gravitational radiation through (detectable) gravitational waves. The pertinent equations of motion are given by Peters (1964), namely,

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1-e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4 \right), \quad (2.25)$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1-e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right), \quad (2.26)$$

with the constant β defined as

$$\beta \equiv \frac{64}{5} \frac{G^3 m M (m + M)}{c^5}. \quad (2.27)$$

Now if one defines the constant c_0 through

$$c_0 = a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304}e_0^2 \right)^{-\frac{870}{2299}}, \quad (2.28)$$

then the time from the second supernova to the conclusion of the inspiral, termed the *coalescence time*, will be given by

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304}e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de. \quad (2.29)$$

The derivations for Eqns(2.25), (2.26) and (2.29) are given in Appendix A. The integral has no closed-form solution, and a numerical integrator is typically used to compute the coalescence time. However, some approximations such as [Mandel \(2021\)](#) do exist and approximate the coalescence time rather well over the entire range of mass ratios.

2.5.1 | Merger rates of compact binaries

The merger rate of a population of compact binaries at time t after formation can be formalised via the heuristic

$$m(t) = \text{number of binaries for which } T_c + T_{\text{rej}} + T_{\text{evo}} < t, \quad (2.30)$$

where T_c is the coalescence time, T_{rej} is the rejuvenation age (the age at which the primary star last accreted mass), and T_{evo} is the evolution time-scale (the length of time the primary spends on the MS). $m(t)$ is called the Delay-Time Distribution (DTD) of a population.

However, the underlying assumption in this is that the population was formed after an instantaneous burst of star-forming activity at time $t = 0$, which is clearly unphysical. We must account for the so-called Star Formation Rate Density (SFRD) $\psi(z)$, assumed to be taken from [Madau & Dickinson \(2014\)](#):

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \text{ M}_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3}, \quad (2.31)$$

where z is the redshift. For the purposes of converting between lookback time t and redshift z , we assume

a standard Λ CDM cosmology with $\Omega_M = 0.3, \Omega_\Lambda = 0.7, \Omega_k = 0, H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Of course, this ignores the effect of mixed-metallicity models, which we account for using the prescription of [Langer & Norman \(2006\)](#), whereupon the SFRD is multiplied by a factor χ :

$$\chi(z, Z) = \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)}, \quad (2.32)$$

where $\hat{\Gamma}$ and Γ are the upper incomplete and complete Gamma functions respectively. This means our Stellar Formation History (SFH) – which takes into account both redshift and metallicity – denoted $\Psi(z, Z)$, is given by

$$\Psi(z, Z) = 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9} \right)^{5.6}} \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)} dz, \quad (2.33)$$

where z_1 and z_2 are the redshifts of our bin edges. The event rate as a function of time, $\mathcal{R}(t)$, is therefore given by the convolution

$$\mathcal{R}(t) = \Psi(z_t) * m(t), \quad (2.34)$$

where z_t is the redshift corresponding to lookback time t , given by the solution to the equation

$$t = t_H \int_0^{z_t} \frac{dz'}{(1+z') \sqrt{\Omega_M (1+z')^3 + \Omega_k (1+z')^2 + \Omega_\Lambda}}, \quad (2.35)$$

with t_H the Hubble time, and $\Omega_M, \Omega_k, \Omega_\Lambda$ the matter density, curvature parameter, and cosmological constant density parameter of the Universe. In general this integral does not have a closed-form solution, however in the case of a flat Universe with $\Omega_k = 0$, an exact solution for redshift⁹ is

$$z = \left(\frac{a}{\sinh(a \sinh(a) + \frac{t}{b})} \right)^{\frac{2}{3}} - 1, \quad (2.36)$$

where $a = \sqrt{\frac{\Omega_\Lambda}{\Omega_M}}$, and $b = \frac{1}{H_0} \frac{2}{3\sqrt{\Omega_M}}$. Detail on how the convolution in Eqn(2.34) is computed may be found in [Richards \(2021\)](#).

2.6 | Numerical implementation

I think there is a world market for maybe five computers.

— THOMAS WATSON, PRESIDENT OF IBM, 1943

Having now discussed the evolution of stars from a theoretical perspective, it becomes imperative to

⁹Integral was computed via SYMPY.

discuss the methods of modelling them. In this section, we provide an overview of the four codes we either use extensively or have developed to carry out the work of this thesis. These four codes are:

1. The Cambridge STARS code (Section 2.6.1), originally developed by Peter Eggleton, which we use to solve the equations of stellar structure and evolution,
2. The TUI code (Section 2.6.2), developed by Jan Eldridge, which performs our binary population synthesis of gravitational wave sources,
3. The KAITIAKI code (Section 2.6.3), developed for this thesis, which is our grid-manager code and allows modification of the input parameters to the Cambridge STARS code programmatically, and
4. The TAKAHE code (Section 4.1.3), developed by Sean Richards for [Richards \(2021\)](#), which evolves the orbits of our binary compact objects.

2.6.1 | The Cambridge STARS code

Brief History

*And through the night, behind the
wheel, the mileage clicking west
I think upon Mackenzie, David Thompson and the rest
Who cracked the mountain ram-
parts and did show a path for me
To race the roaring Fraser to the sea.*

—“NORTHWEST PASSAGE”, BY STAN ROGERS

In 1959, Louis George Henyey published his method for automatic stellar evolution ([Henyey et al., 1959](#)). Prior to this – and the widespread availability of the computer – a model was treated as a mathematical process and thus was computed, often labouriously, by hand. For instance, in [Eggleton \(1967\)](#), Eggleton writes that

Since we have seen in Section 3 that the structure of the flame zone can be obtained independently (almost) of the rest of the star, it is reasonable to expect that [thermal] instabilities of the type considered by Schwarzschild & Härm can also be examined in this way, so that the considerable labour of computing the structure of an entire star can be avoided.

– [Eggleton \(1967\)](#)

The Cambridge STARS code was one of the earliest detailed stellar evolution codes to be written, though initially it only modelled one star in detail. To model a binary system, tweaks were made whereupon the

primary would be modelled in detail whilst approximating the secondary with the equations of [Hurley et al. \(2002\)](#) until the primary supernova, at which point the secondary would be evolved in detail. This had its merits; notably computation time was saved, and the results were generally accurate anyway.

However, [Stanciliffe & Eldridge \(2009\)](#) introduced detailed modelling of *both* stars in a binary system into the STARS code – that is, the interiors of both stars were computed in detail, alongside the simultaneous computation of the orbital parameters. Since then, a wide array of improvements have been made to the numerical stability of the code, as well as introducing or refining new physics.

There are four variants of this code:

1. The Cambridge STARS code, which is the version originally developed by Peter Eggleton and originally described in [Eggleton \(1971\)](#). This version was originally only capable of solving the equations for a single star.
2. The ELDRIDGE STARS code described in [Eldridge & Tout \(2004b,a\)](#); [Eldridge et al. \(2008, 2017\)](#), which is the version used to generate the fiducial BPASS v2 grids. This version models the primary in detail while approximating the secondary using the equations of [Hurley et al. \(2002\)](#).
3. The BSTARS code, which is the unmodified version lately available from Cambridge, described in [Stanciliffe & Eldridge \(2009\)](#). This version models both stars in detail throughout their lives.
4. The AOTEAROA STARS code, which is the version of the code incorporating the changes we have made.

Unless context otherwise requires, when we refer to “the STARS code” with no further qualification, we are referring specifically to the AOTEAROA STARS code.

Matrix Inversion

The heart of STARS is a matrix inversion algorithm. Although computing the inverse of a matrix is a polynomial time process, it can still take considerable time to invert. STARS implements a Henyey matrix ([Henyey et al., 1964](#)), which is of size $n = (4 + n_{\text{comp}})N_{\text{mesh}}$, with n_{comp} the number of equations to be solved and N_{mesh} the number of meshpoints in the model. An individual element of the Henyey matrix is given by

$$\mathbf{H}_{ij} = \frac{\partial (\text{equation } i)}{\partial (\text{variable } j)}. \quad (2.37)$$

For a “typical” binary computation, with $n_{\text{comp}} = 30$ equations and $N_{\text{mesh}} = 199$ meshpoints, the matrix is already 6766×6766 , for a binary which evolves up the AGB (the theoretically most complex calculation we perform) with $N_{\text{mesh}} = 999$, the matrix is 33966×33966 . Introducing k new equations into the code causes the runtime of the matrix inversion calculation to scale like $\mathcal{O}(n^3 + k^3 N_{\text{mesh}}^3)$.

The matrix inversion is simple Gauss-Jordan inversion, which is of complexity class (for an $n \times n$ matrix) $\mathcal{O}(n^3)$. [Coppersmith & Winograd \(1987\)](#) showed that this could be optimised to $\mathcal{O}(n^{2.376})$ and conjectured reducibility to $\mathcal{O}(n^2)$.¹⁰ Since our code is embarrassingly parallelisable, and can be executed in parallel simply

¹⁰The current “best” optimisation is $\mathcal{O}(n^{2.3727})$ ([Williams, 2012](#)). DeepMind’s AlphaTensor has claimed a faster result in [Fawzi et al. \(2022\)](#), though they do not state a complexity class. This is because “...this does not yield practical algorithms.”

by increasing the number of cores available, the payoff for decreasing the runtime by one polynomial order is inconsequential, and thus we use the straightforward implementation.

Initial Condition Selection

In this section we provide an account for the selection of our general initial conditions. Unless otherwise noted in this thesis, it is assumed that the values will be using the parameters described here and in Appendix B.

In our implementation of Mixing Length Theory (MLT) (Vitense, 1953; Böhm-Vitense, 1958; Cox & Giuli, 1968), we use an MLT parameter of $\alpha_{\text{MLT}} = 1.985$, selected via calibration against a model of the Sun. This is effectively a dissipativity parameter, and represents the depth at which a parcel of matter penetrates into a lower layer before becoming fully dissipated. The value was selected by modelling a $1 M_{\odot}$ star and varying only the mixing length parameter until it had a luminosity of $1 L_{\odot}$ and a radius of $1 R_{\odot}$, at the present age of the Sun.

Mesh spacing

A natural question that arises when modelling a star is where internally to model it – there is a counterbalance between the level of detail and the time cost of the matrix inversion. The obvious answer would be to distribute a mesh equally spaced within radius of the star, however the nonhomogeneity of a star, plus the potential for very thin shells of physically important material, necessitate a more intelligent approach.

Instead of being equally spaced *radially*, these mesh points are taken to be equally spaced within a function $\mathcal{Q}$, which is taken from Stancliffe et al. (2004) and given for a binary that does not have AGB-specific treatment enabled by

$$\begin{aligned} \mathcal{Q}(P, T, M, R) = & c_4 \ln P + c_5 \ln (P + c_{\text{H}}) + c_2 \ln (P + 3P_{\text{He}}) + c_2 \ln (P + 0.3P_{\text{He}}) + c_7 \ln \left(\frac{T}{T + 2 \times 10^4} \right) \\ & + \ln \left(\frac{c_6 M^{\frac{2}{3}}}{M + M^{\frac{2}{3}}} \right) - c_3 \ln \left(\frac{R^2}{c_8} + 1 \right), \end{aligned} \quad (2.38)$$

where P, T, M, R are the total pressures, temperatures, masses, and radii of each star, the constants c_i are user controllable and given in Appendix B, P_{He} is the pressure in the helium-burning shell, and

$$c_{\text{H}} = \begin{cases} 0.1P_{\text{H}} & X_{\text{H}} \geq 0.15 \\ 0.1\mathcal{P} & \text{otherwise} \end{cases} \quad (2.39)$$

where P_{H} is the pressures in the hydrogen-burning shell. Here, $\mathcal{P}$ represents the pressure in the hydrogen-burning shell when $X_{\text{H}} = 0.15$, and its inclusion obviates numerical issues if the envelope is removed and there is no hydrogen burning occurring. The choice of mesh spacing coefficients $c_1 \dots c_{10}$ leads to different areas of the star being prioritised. Coordinates c_1, c_9 , and c_{10} are not used except in AGB-specific evolution, which is not required for most cases (and requires a change of parameterisation in the pressure-shell terms). During

binary evolution, c_5 is hardcoded to be 0.15, and c_2 to be 0.0, to focus meshpoints into the hydrogen-burning shell.

Placement of the mesh is controlled by the derivative $\frac{\partial \mathcal{Q}}{\partial k}$, where k is the meshpoint number (Stancliffe, 2016). Specifically, meshpoints are placed at locations where

$$\left. \frac{\partial \mathcal{Q}}{\partial k} \right|_k - \left. \frac{\partial \mathcal{Q}}{\partial k} \right|_{k+1} = 0. \quad (2.40)$$

The mesh is thus adaptive (insofar as it changes to place meshpoints where they are needed for evolution), non-Lagrangian (insofar as the derivative is not at constant mass coordinate), and inviscid (insofar as the gradient of $\mathcal{Q}$ is constant through meshpoint number). This combination of requirements is sometimes referred to as an Eggletonian mesh. If one desires to make the mesh viscous, one can follow Stancliffe et al. (2004) and write, for $0 \leq \gamma \leq 1$,

$$\left(\left. \frac{\partial \mathcal{Q}}{\partial k} \right|_k - \left. \frac{\partial \mathcal{Q}}{\partial k} \right|_{k+1} \right) (1 - \gamma) + \gamma \left(\left. \frac{\partial m}{\partial t} \right|_k \right) = 0, \quad (2.41)$$

where now the parameter γ is a weighting coefficient such that if $\gamma = 1$, then the mesh spacing function is irrelevant and the parameters are spaced equally in mass, and if $\gamma = 0$, the original mesh spacing function is used. In between, if $\gamma = \gamma(k, dt)$, then parts of the mesh are allowed to be Lagrangian and parts non-Lagrangian. We typically keep the mesh non-viscous and non-Lagrangian, as a viscous mesh is most useful for AGB evolution and evolving through thermal pulses.

More detail is provided in Eggleton (1971) for the original mesh spacing function, and Stancliffe et al. (2004) for the binary mesh spacing function.

Timestep control

The STARS code utilises an adaptive timestep control mechanism, whereby the timestep is allowed to shrink to model rapid changes or grow if the star is broadly inert and unchanging. The core computation is a parameter Δ , given by

$$\Delta = \frac{d_{\text{opt}}}{\sum_i \sum_j |\delta x_{ij}|}, \quad (2.42)$$

where d_{opt} is a pre-defined target timestep (typically taken to be ~ 1 , unless otherwise specified; see Appendix B for our selections), and δx_{ij} is the change in variable x_i at meshpoint j . In this calculation, luminosity is explicitly excluded as this would cause the timestep to explode during certain points of the star's lifetime. A new timestep, d' , is selected from the previous timestep d by

$$d' = \begin{cases} \min[d_1, \Delta] d & \Delta < 1 \\ d & \Delta = 1 \\ \max[d_2, \Delta] d & \Delta > 1 \end{cases}, \quad (2.43)$$

where d_1 and d_2 are pre-determined control parameters. Most commonly we take $d_1 = 0.1$ and $d_2 = 1.2$ unless otherwise specified.

Numerical instabilities

Stellar modelling is a nontrivial endeavour and there are many cases where the code can fail seemingly arbitrarily. The largest cause of numerical instability is the helium flash, which occurs in low-mass stars between about $0.8 M_\odot \lesssim M_{\text{ZAMS}} \lesssim 2.5 M_\odot$. In this band, the triple- α process occurs within a highly electron-degenerate helium core. As a result, the temperature of the star increases sharply during this process, leading to thermal runaway. This temperature increase in turn results in an increase in the luminosity (which peaks at $\sim 10^{10} L_\odot$). The flash itself – being the period of thermal runaway – occurs on the order of a few minutes, though the spike in luminosity lasts approximately 7 Myr (for the $2.51 M_\odot$ star shown in Figure 2.13).

A method termed “pseudoevolution” can be used to overcome this source of error. Briefly, a complete model (termed the *reference model*) of a $3 M_\odot$ star is evolved with a full model description saved at every timestep. When the helium flash occurs in a model (heuristically, $0.1 \leq M_{\text{He}}/M_\odot \leq 0.55$ and $\log T_c/\text{K} > 7.9$), we query the reference model for a model with the same helium core mass as the failed model. Using a custom subroutine in STARS, we turn off all nuclear and thermal burning and remove mass from the reference model to match the mass of the failed model. This model is then allowed to evolve as per normal, having overcome the helium flash. We note that this is not a *perfect* method, as this does induce discontinuities in the plots, as shown in Figure 2.14, though these are essentially insignificant.

The other stage where low-mass stars are prone to fail is during a thermal pulse (Schwarzschild & Härm, 1965, 1967). The mass range for thermal pulses is broader, from $\sim 0.8 M_\odot$ to $\sim 8 M_\odot$. A thermal pulse occurs as a star ascends the AGB; as a CO core or ONe core has developed, shells of burning helium and hydrogen (separated by a region of almost pure helium) sit atop the core (Pols et al., 2001). As the intershell region dissipates due to dredge up or helium burning, hydrogen burning reignites in the hydrogen shell. As both shells burn (at the same rate), the helium-burning shell becomes unstable and eventually a process of thermal runaway begins due to the instability of a thin shell. This causes a spike in the luminosity, termed a thermal pulse. This can persist for many pulses – Stancliffe (2006) observes pulses before model termination, while Stancliffe (2005) observes 17 pulses for a model similar to those in Figure 2.13.

The method to overcome this numerical instability is involved and not fully implemented in STARS. Briefly, the number of meshpoints must be drastically increases near the onset of thermal pulses; up to 999 instead of the typical 199/499. The timestep must also be forcibly decreased; thermal pulses typically last for around a year (Pols et al., 2001), and up to $\sim 10^2$ yr (Karakas & Lattanzio, 2014), so the timestep must be reduced to not miss them (and to not have the modulus of the change in parameters – $\sum_i \sum_j |\delta x_{ij}|$ in Eqn(2.42) – and thus the error, increase too much). This method was implemented to generate Figure 2.13, though only two pulses were achievable.

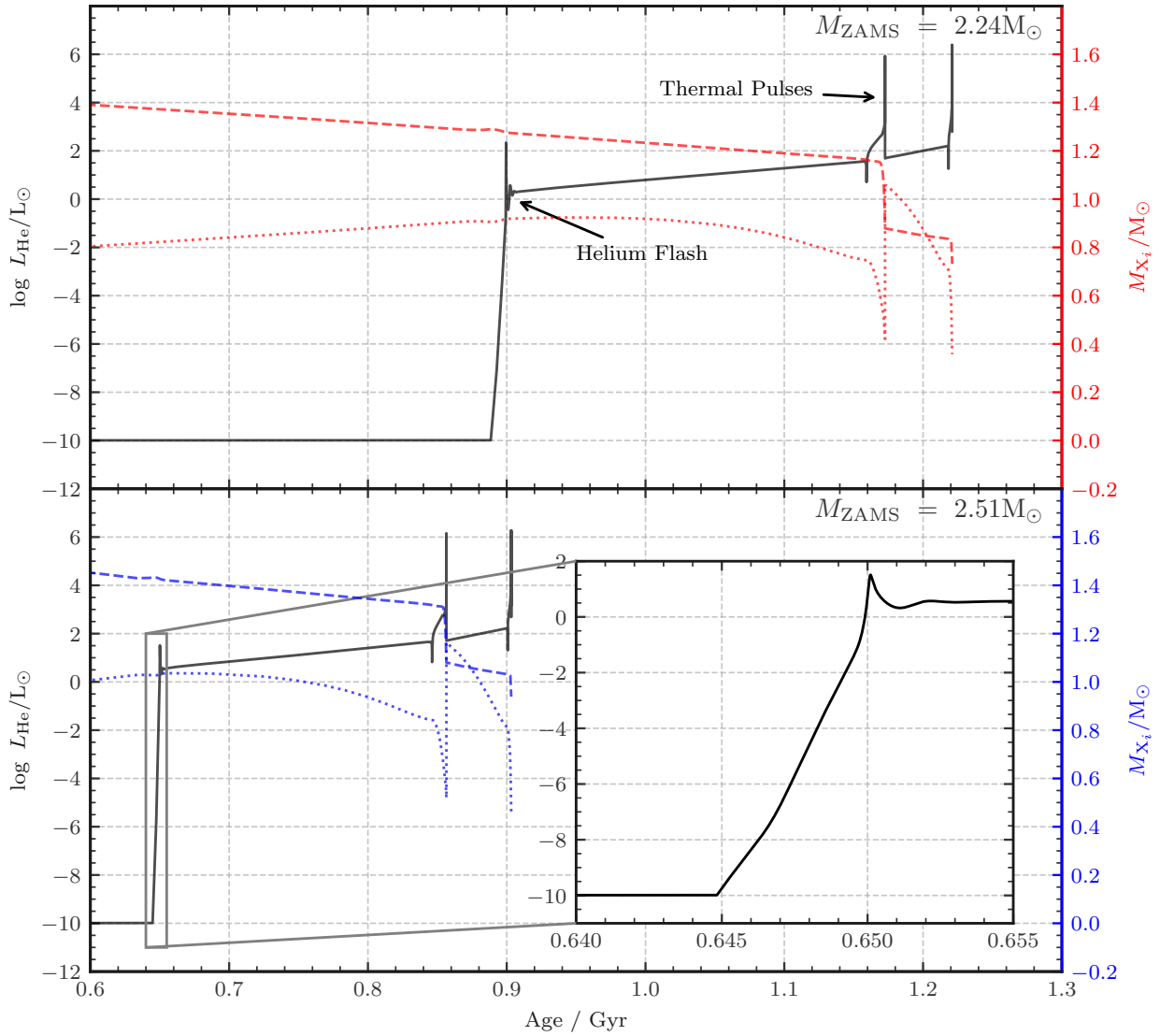


Figure 2.13: Two stars of a similar mass that undergo a helium flash and two thermal pulses. Shown in the inset is a zoomed-in helium flash that lasts ~ 7 Myr. Both stars undergo two thermal pulses before the code crashes due to the relative error in the calculation increasing too rapidly. The two stars could, in principle, undergo many more thermal pulses before their deaths.

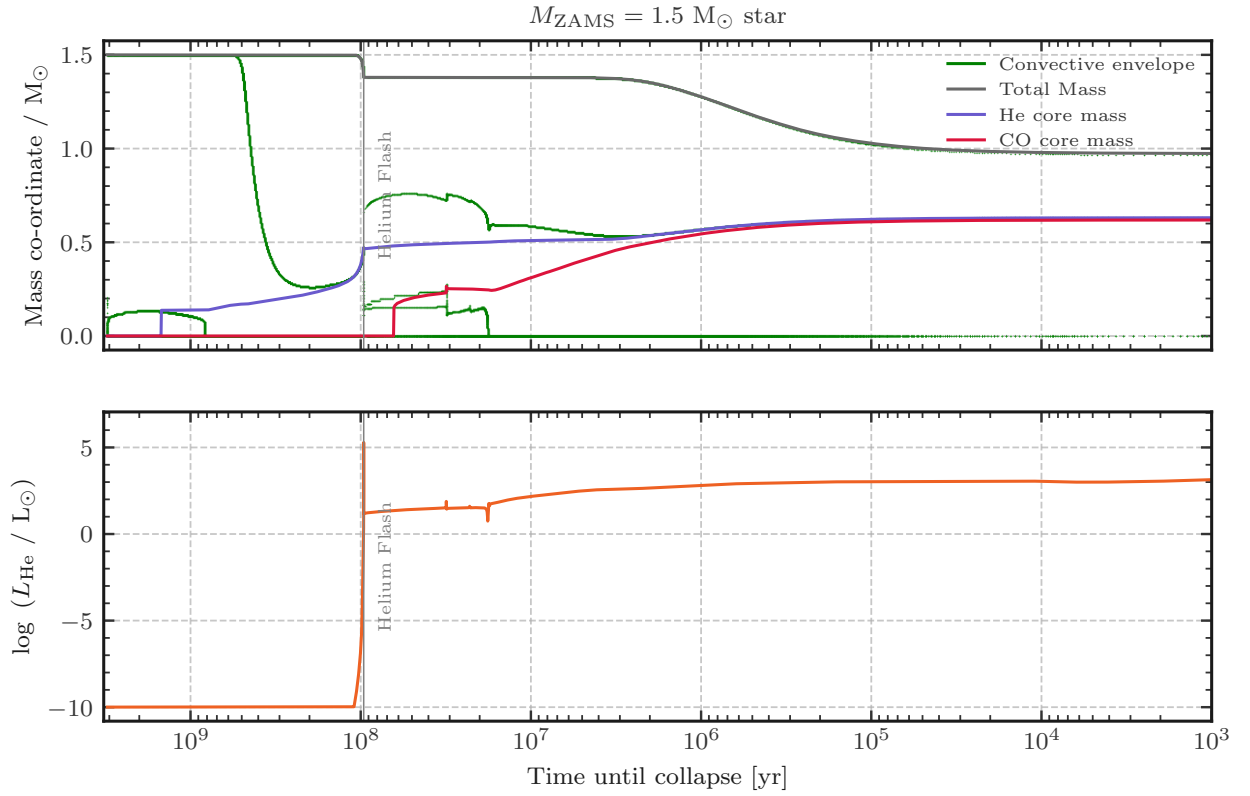


Figure 2.14: The helium flash (gray line) shown on a Kippenhahn diagram. The total mass is shown as the solid gray line, and the CO core mass is shown in red. The green lines show the boundaries of the convective regions of the star.

Critical modifications

The major modifications we have made to the STARS code are as follows.

Firstly, the floating-point precision has been increased to double precision for all variables. This has minor effects on the evolved models, insofar as they are generally able to evolve to slightly further stages. This step has previously been undertaken, however not all variables were correctly converted. This did not invalidate any models previous versions of the STARS code has generated; rather it means they terminated slightly earlier than the point to which they may have been able to evolve.

As part of this, we also undertook an extensive refactor of the codebase. Whilst some of the code remains as originally written, it is now more compliant with modern programming practices and is in some sense easier to alter with new routines.

We have also improved mass loss interpolation in the MASSLOSS.F routine. Whilst not changing the underlying assumption that mass transfer is governed by [de Jager et al. \(1988\)](#); [Vink et al. \(2001\)](#) (Section 2.4.3), we have improved the transition between typical O-type mass loss and WR mass loss rates by slowing turning on the WR mass loss rate as the surface hydrogen abundance drops below 0.4. The transition was previously abrupt and would cause the code to fail due to numerical instability if a non-WR star entered this regime.

Similarly to this, the code now manually reduces the timestep in the run-up to, during, and after the contact binary phase. Evolution during this phase breaks thermal equilibrium (i.e., the star evolves on a shorter timescale than its Kelvin-Helmholtz time-scale) and hence the evolution can change rapidly as the

star attempts to adjust back to thermal equilibrium. The contact phase, for example, typically last for between 100-200 kyr in our models.

We have also begun implementing an $\alpha - \lambda$ treatment of CEE, though we stress that this has not been rigorously tested yet, and is thus disabled for the majority of our models.

2.6.2 | The TUI code

TUI is the core population synthesis code within the BPASS project. TUI takes as input the output from the STARS code. TUI was first described in [Ghodla et al. \(2023\)](#), and was further described in [Stevance et al. \(2023\)](#), though we note that their version is technically a refactor of the version we use. The predominant function we use TUI for is the matching of pre- and post- supernova models for Chapter 4, and in this section we provide a brief account of that functionality of TUI (though the control of the population synthesis models is beyond the scope of this thesis).

TUI has precompiled “input_” files which report a STARS plot file, an IMF weight, and a model type (merger/normal primary/normal secondary/single model/quasi-homogeneous evolution binary/single star). After loading in a model, TUI applies a given kick and computes the new orbital parameters from [Tauris & Takens \(1998\)](#). Crucially, a core assumption of BPASS is that the orbit remains circular after the first supernova and is allowed to be eccentric after the second. Of course, nature cares little for our assumptions, and the supernova may cause the binary to become eccentric after the first supernova; TUI accounts for this by setting the separation after the first supernova to be the semi-latus rectum of the eccentric binary. TUI repeats this process after the second supernova.

For the purposes of Chapter 4, if the binary remains bound (has $0 \leq e < 1$) after the second supernova, we record the following parameters:

1. The mass of the primary,
2. The mass of the secondary,
3. The orbital separation,
4. The eccentricity,
5. A weight term proportional to the IMF weighting,
6. The rejuvenation age, which is the time since the companion last accreted material, if it accreted more than five per cent of its initial mass (see [Richards et al., 2023](#)), and
7. The evolution age, which is the length of time from the secondary’s ZAMS until its death.

This provides us with the post-secondary supernova orbital parameters which determine the properties of the merger, such as the event rate $\mathcal{R}(z)$ and its path through period-eccentricity space.

2.6.3 | The KAITIAKI grid-watcher code

KAITIAKI is a grid-watcher code we have developed that sits atop the STARS code and manages its execution. It was developed primarily to manipulate the fixed-width text files that STARS uses, in contrast to the “typical” manipulation done through the command line. However development proceeded in such a way that it now provides an interface to all of the (relevant to this thesis) output and input files. Further, it also now allows the user to control jobs through a PYTHON interface, greatly simplifying grid calculations.

The core of KAITIAKI is the `STARSController()` interface. The most naïve run would be to initialise a model of, say, $8 M_{\odot}$ and evolve it until it evolves up the AGB. A sample of code for this is shown in Code Sample 2.1 which evolves a $10 M_{\odot}$ star.

Code Sample 2.1: A KAITIAKI script to evolve a $10 M_{\odot}$ star

```

1  import kaitiaki
2
3  # First, instantiate the STARS code. run_bs is the location of the run_bs
4  # bash script, which sets up the symlinks and actually runs the code.
5  STARS = kaitiaki.STARS.STARSController(run_bs='..')
6
7  # Loads the default data file and solar metallicity opacity table
8  STARS.blit('z020')
9
10 # Loads a 1 Msun model
11 STARS.load_default_modin()
12
13 # Sets the code up to target a ZAMS mass of 10 Msun.
14 STARS.setup_zams_inflation(10)
15
16 # Executes the code, such that the final modout will be a 10 Msun model.
17 STARS.run()
18
19 # Equivalent to "tail -399 modout > modin" -- transfers the last 2*199+1
20 # lines from modout (the last converged model) to modin
21 STARS.modout_to_modin()
22
23 # Again, to reload data
24 STARS.blit()
25
26 # Set up the key parameters for our run
27 STARS.configure_parameters({
28     'DT1': 0.1, # Fraction of timestep to allow for lower variation
29                # (if the timestep is cut, it can be cut to 10% of its
30                # present value)
31     'DDD': 2,   # Modulus of total desired timestep
32     'IML1': 5,  # WR mass-loss rates
33     'RML': 0,   # This was set to 10 Msun during inflation -- should be
34                # zero so no mass is added except via the usual ML channels
35                # (this parameter, formally, sets a constant ML rate so
36                # that in the absence of other mechanisms, the mass would
37                # always decrease)
38     'IMODE': 1, # Single star mode
39     'IX': 1,   # Hydrogen burning
40     'IY': 1,   # Helium burning
41     'IZ': 1,   # Metal burning
42     'ITH': 1,  # Thermal energy generation
43     'ISTART': 1 # Reset age, NMOD, and the timestep (to 10% of T_KH)

```

```
44 })  
45  
46 # Execute the model, run until the AGB.  
47 STARS.run()
```

2.7 | Chapter Summary

*If in the first act you have hung a pistol
on the wall, then in the following one it
should be fired. Otherwise don't put it there.*

— ANTON CHEKHOV

In this Chapter, we have explored the evolution of stars from pre-MS evolution to supernova, accounting for the effects of binary interaction. This represents the state of the field as at the commencement of this work. In future chapters we will detail the relevant changes we have made to these prescriptions, motivating the work we have performed.